How To Untie Your Mobile

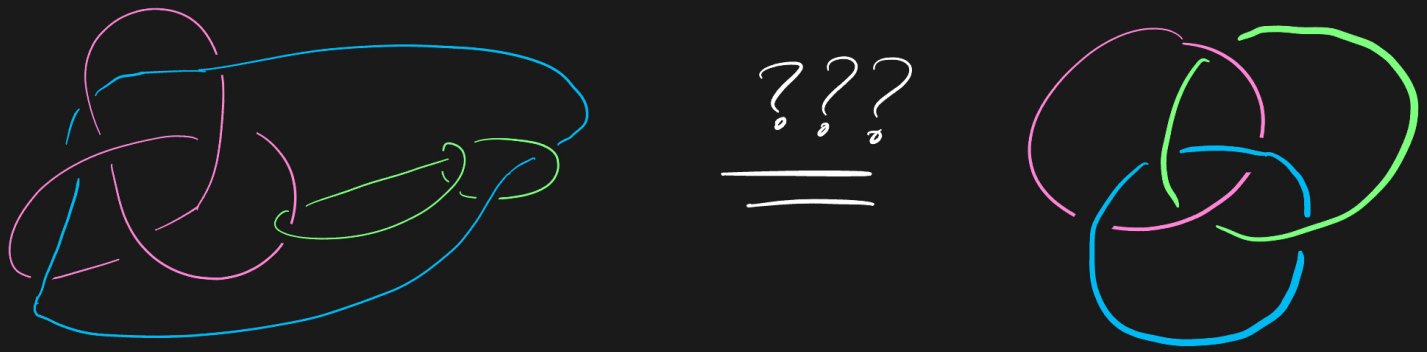
Gabe Field

10/Aug/2026

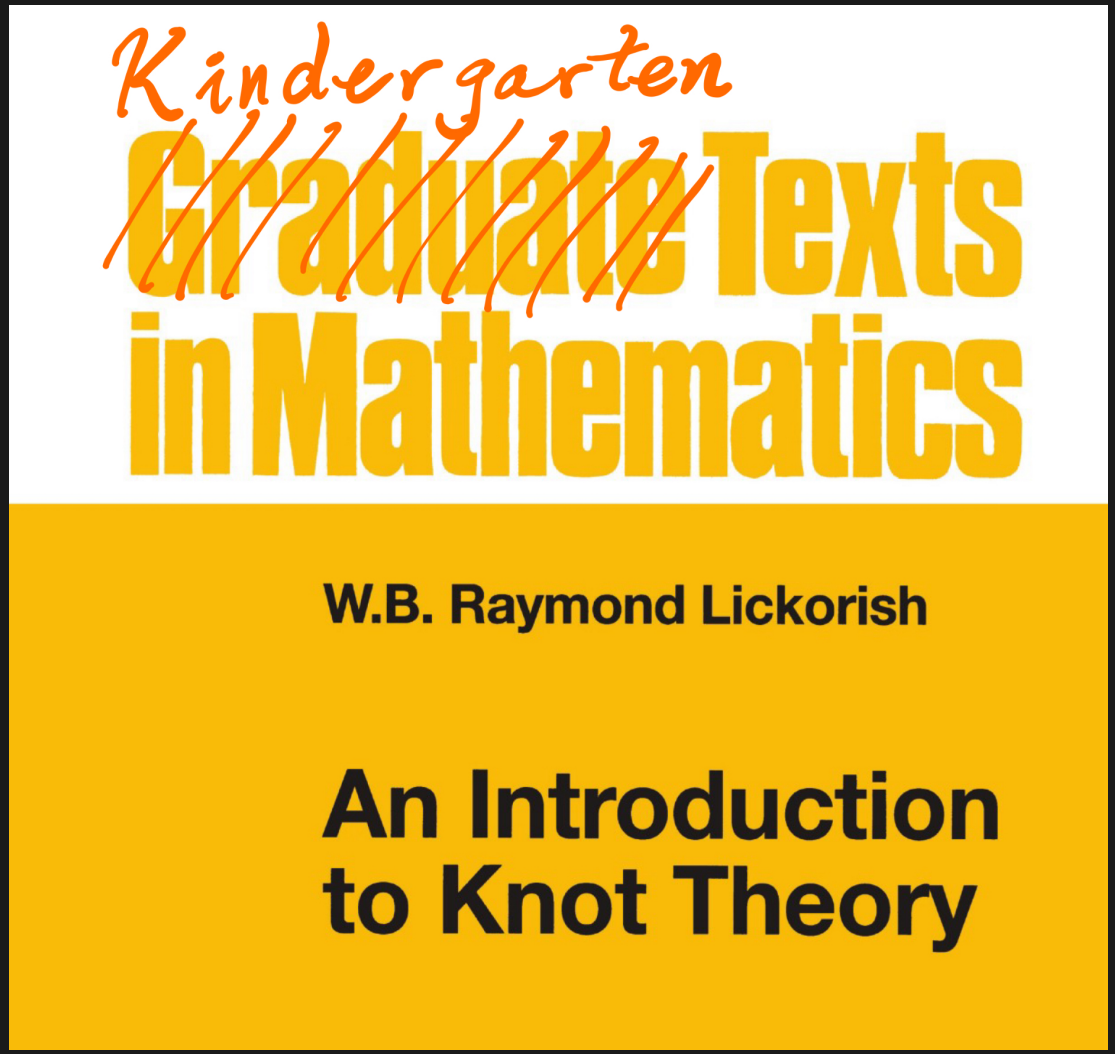
Thanks!

Thanks to *Daniele*, *Christian*, *Mohammadali*, *Luke* and *Aaron* for helping me prepare this talk.

What is knot theory all about?



Frustrations



Where are we going today?

Goals:

- **Main:** Tell knots apart (hard)

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- **Main:** Tell knots apart (hard)
- **Today:** Develop a tool for *developing tools* to tell knots apart.

Outline

1. The Polygonal Perspective
2. The Smooth Perspective
3. Summary

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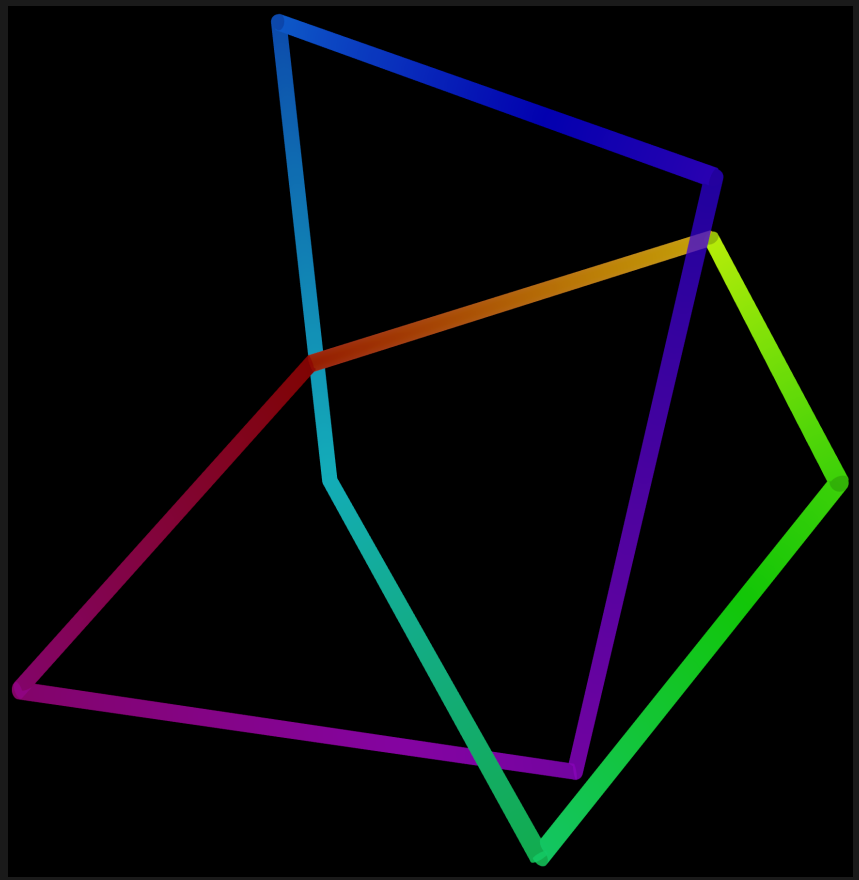
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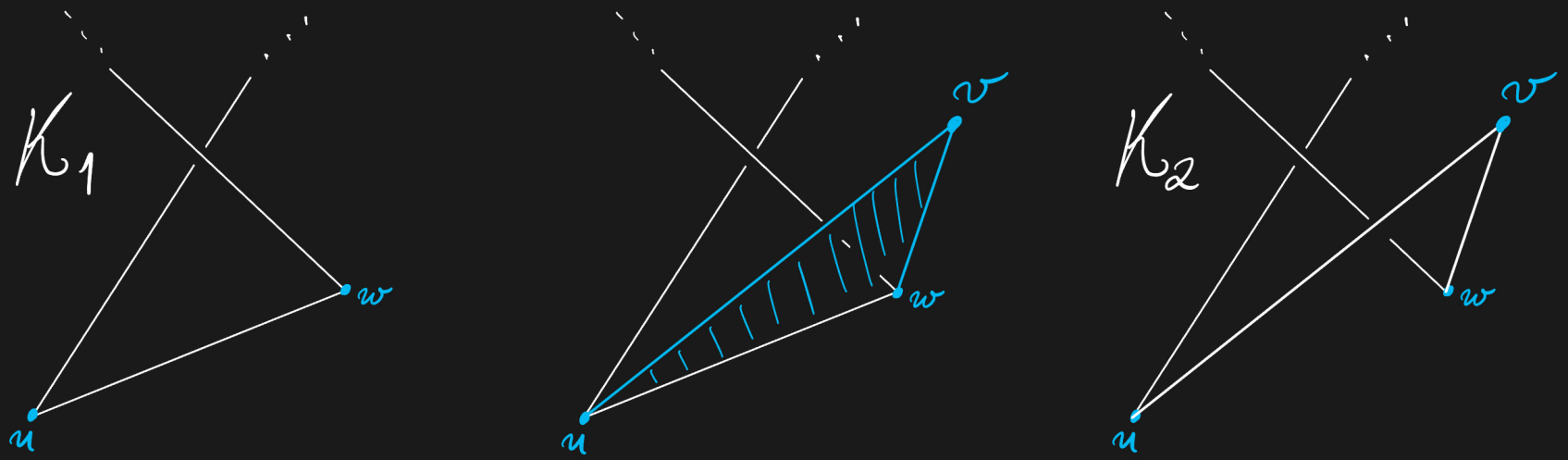
What are knots?

Def. A *knot* is a piecewise linear simple closed curve in $\mathbb{R}^3$.



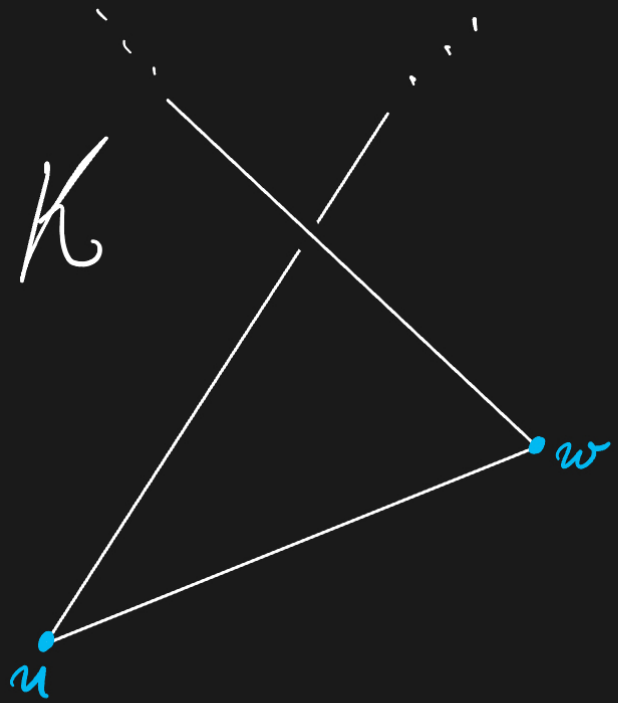
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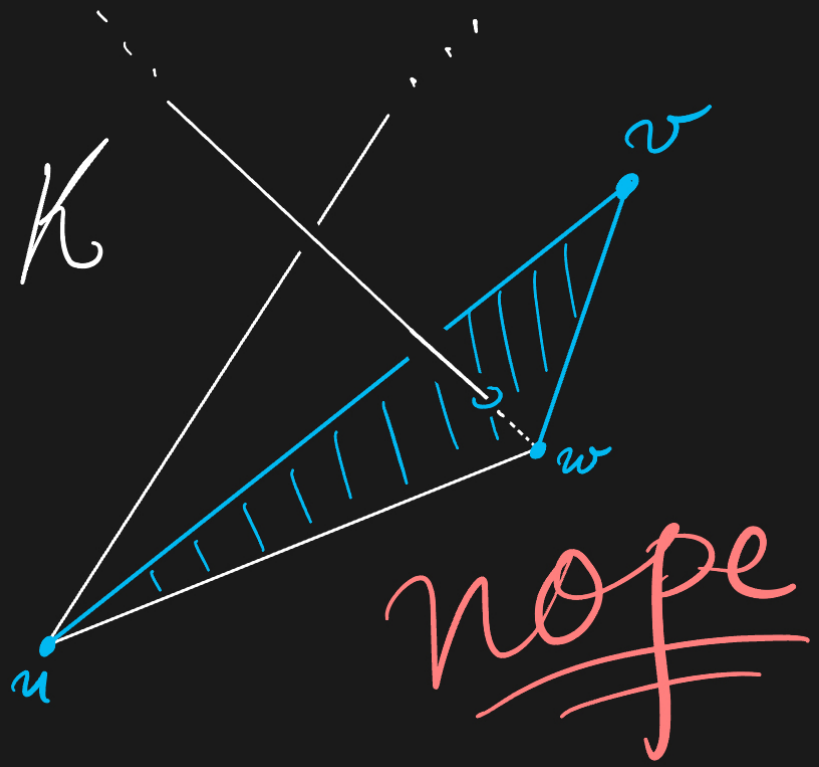
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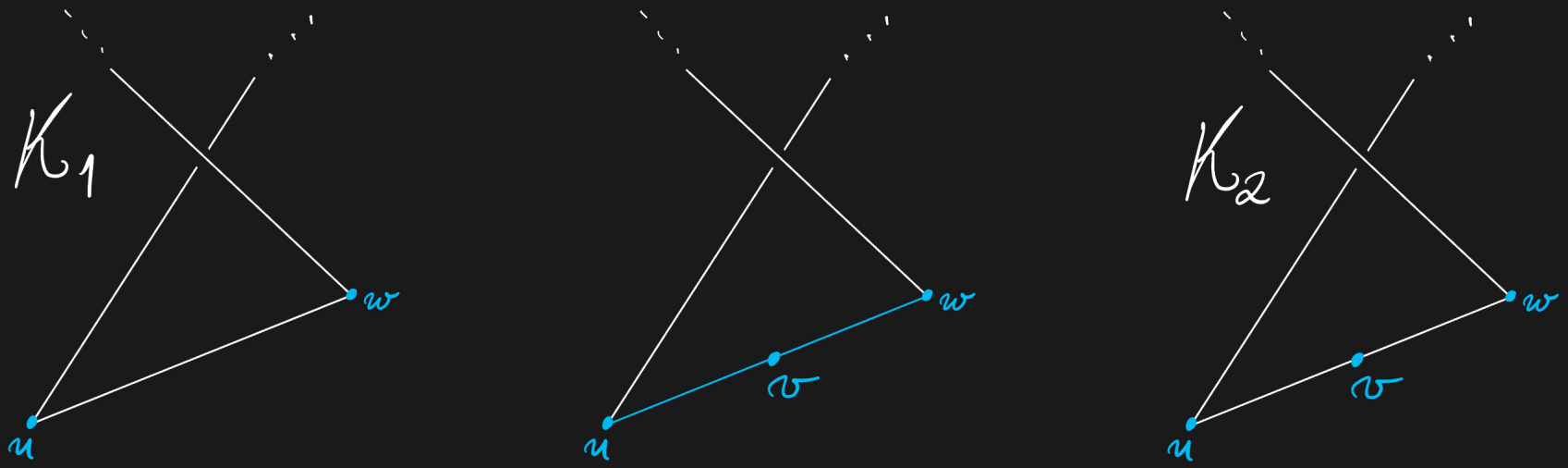
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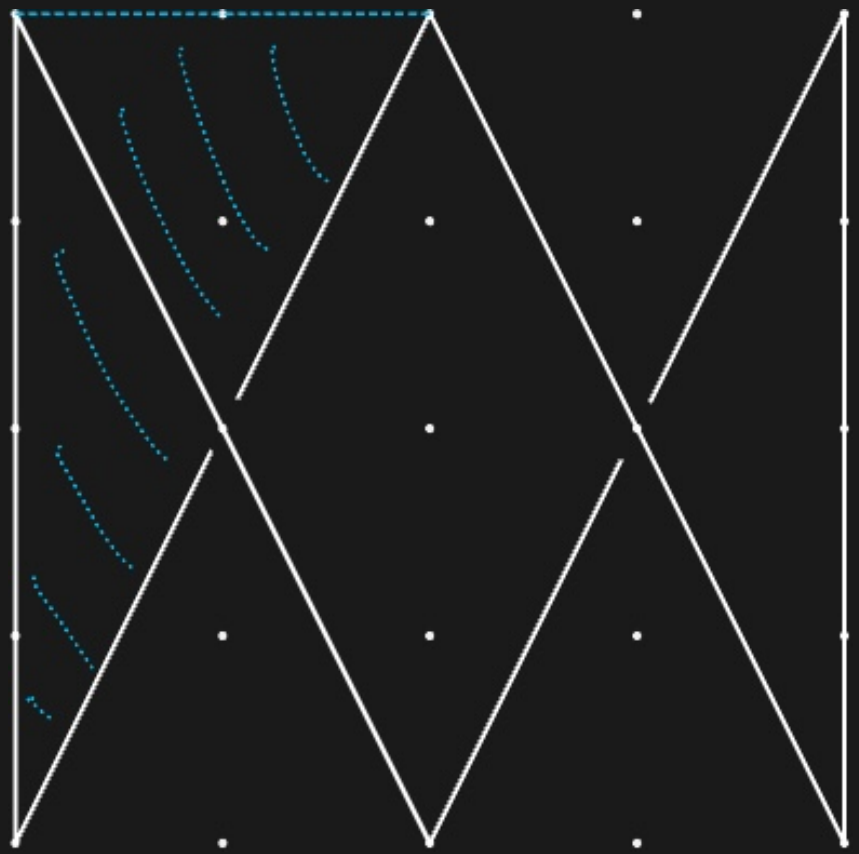
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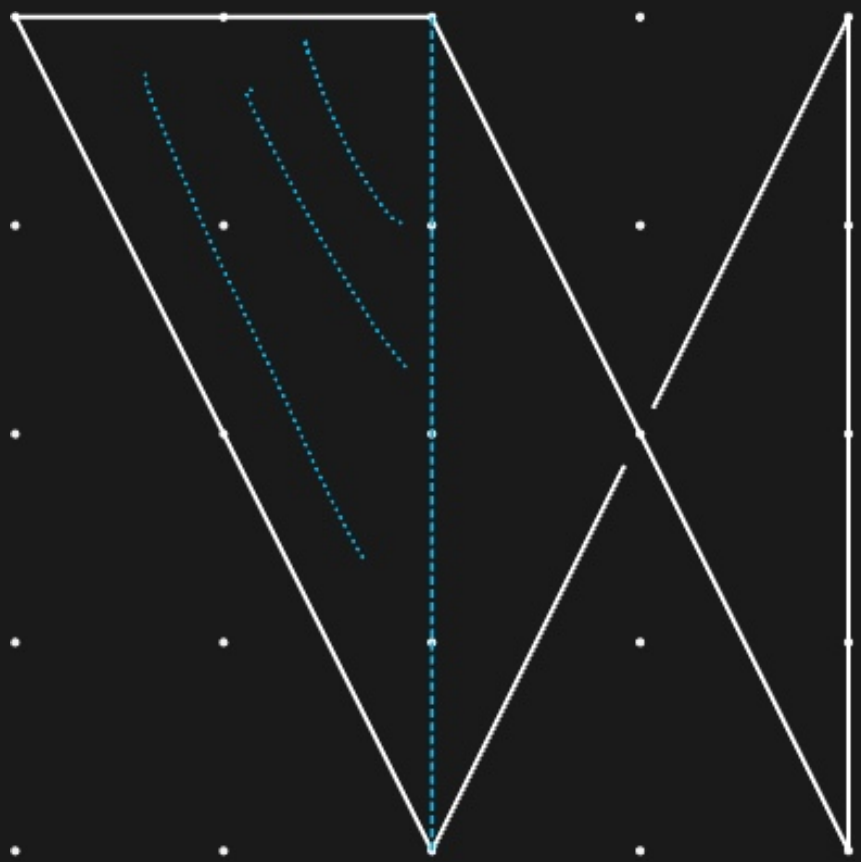
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Def. Two knots are *isotopic* (by moves) just when they are related by a sequence of Δ- and Δ^{-1}-moves.



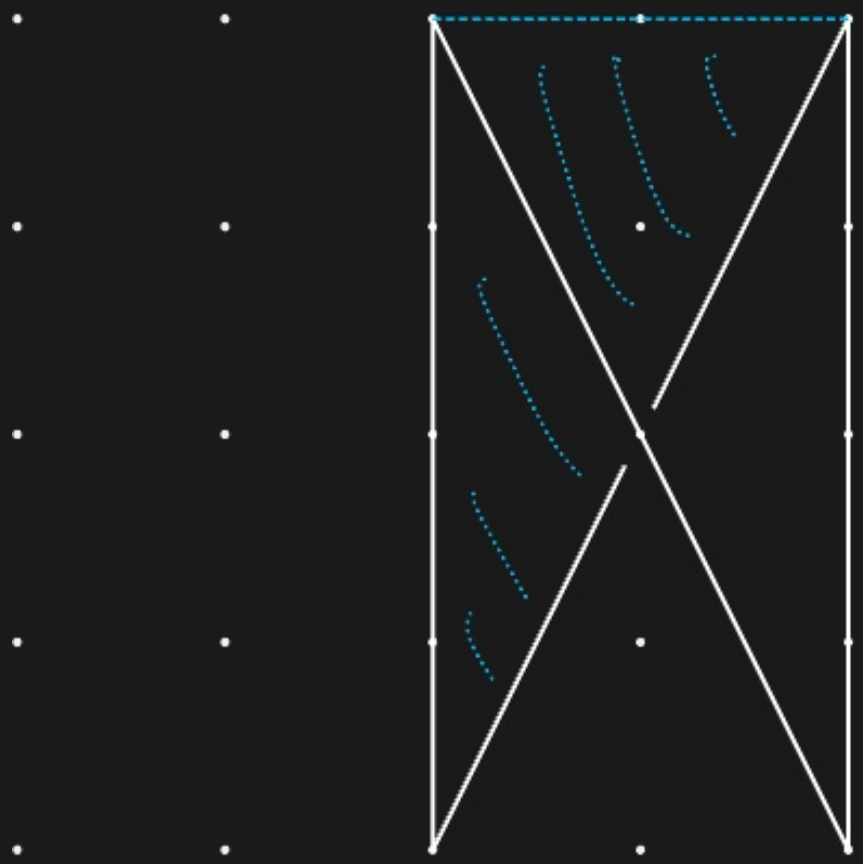
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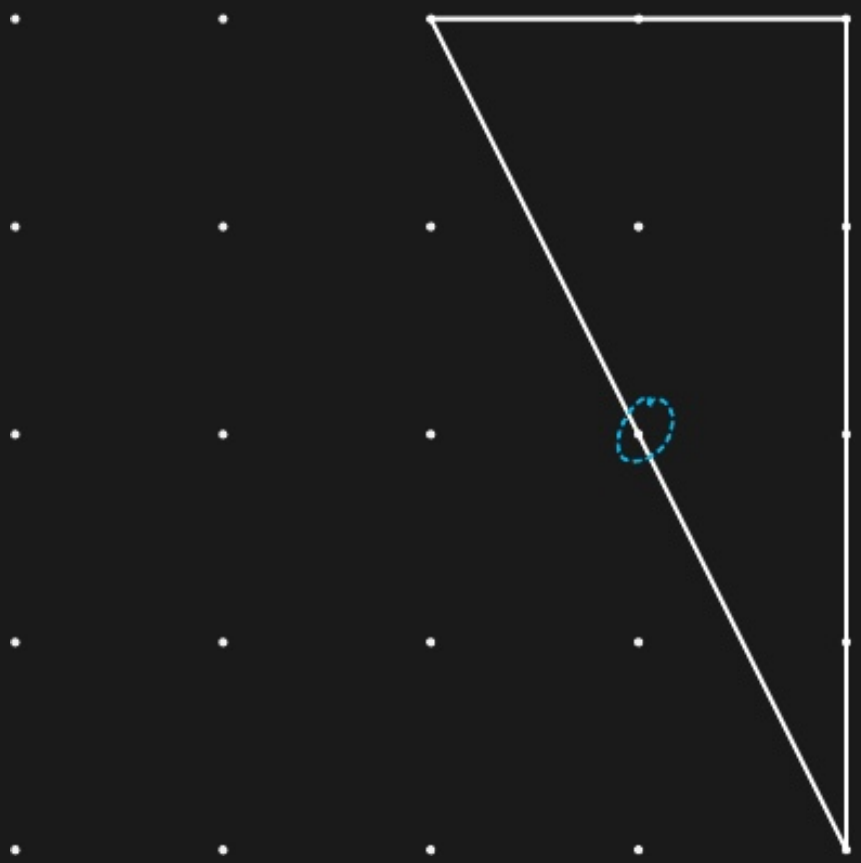
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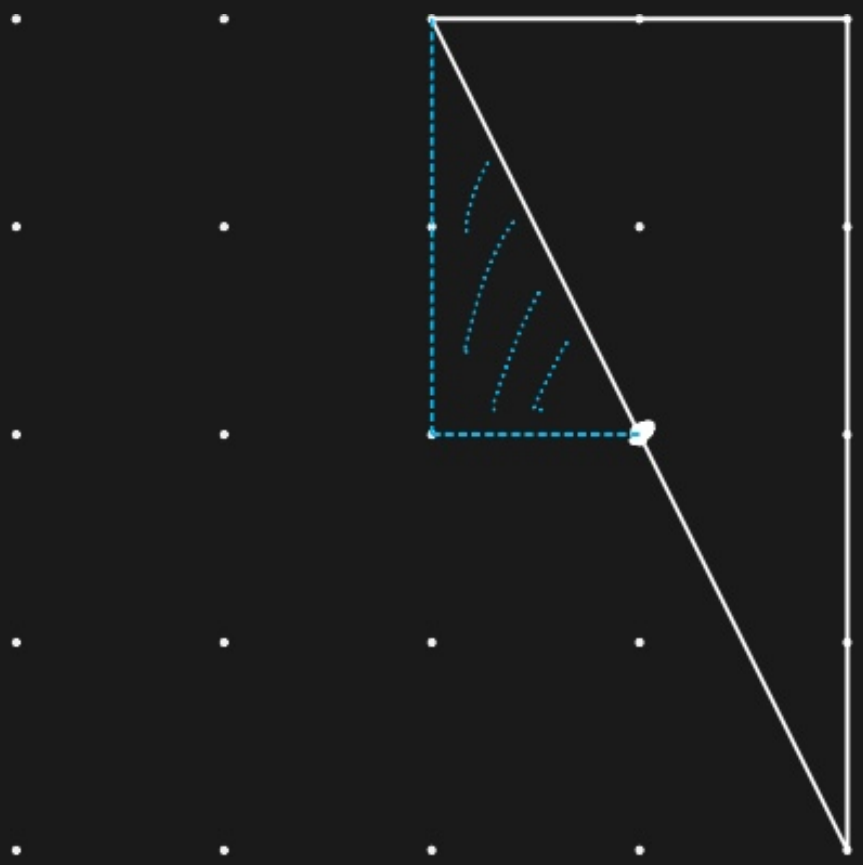
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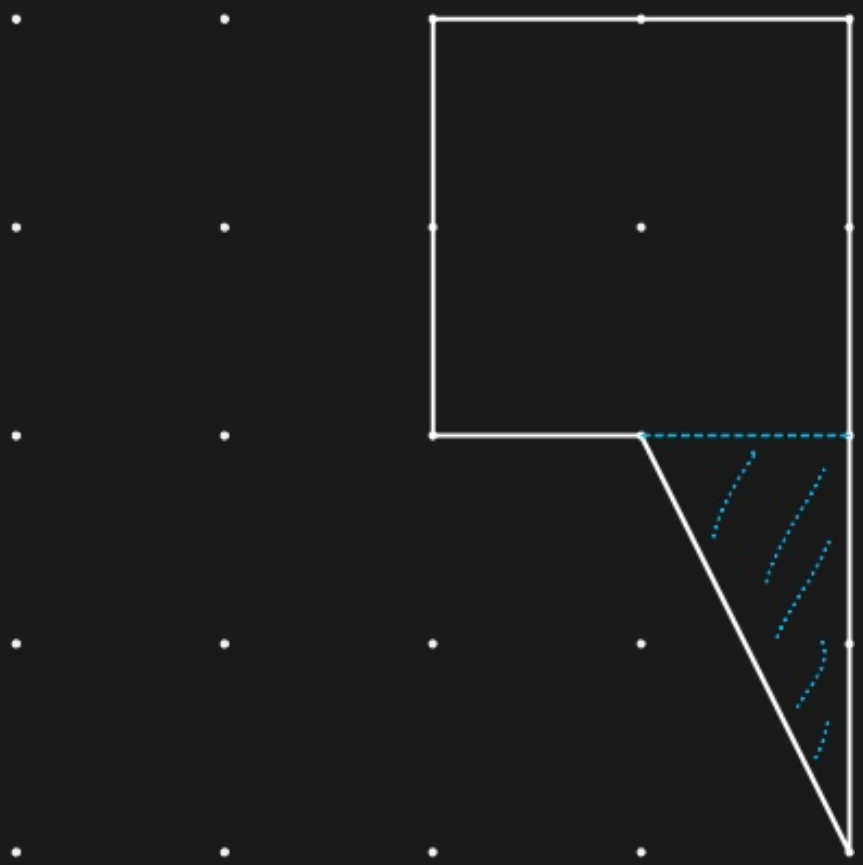
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What are Knots?

Def. A *knot* refers to either a curve as before, or a class of them modulo isotopy.

Question. How do we tell knots apart?

Knot invariants

Def. A *knot invariant* is a function out of the set of knots modulo isotopy.

$$f(K_1) \neq f(K_2) \quad \Longrightarrow \quad K_1 \neq K_2$$

Knot invariants

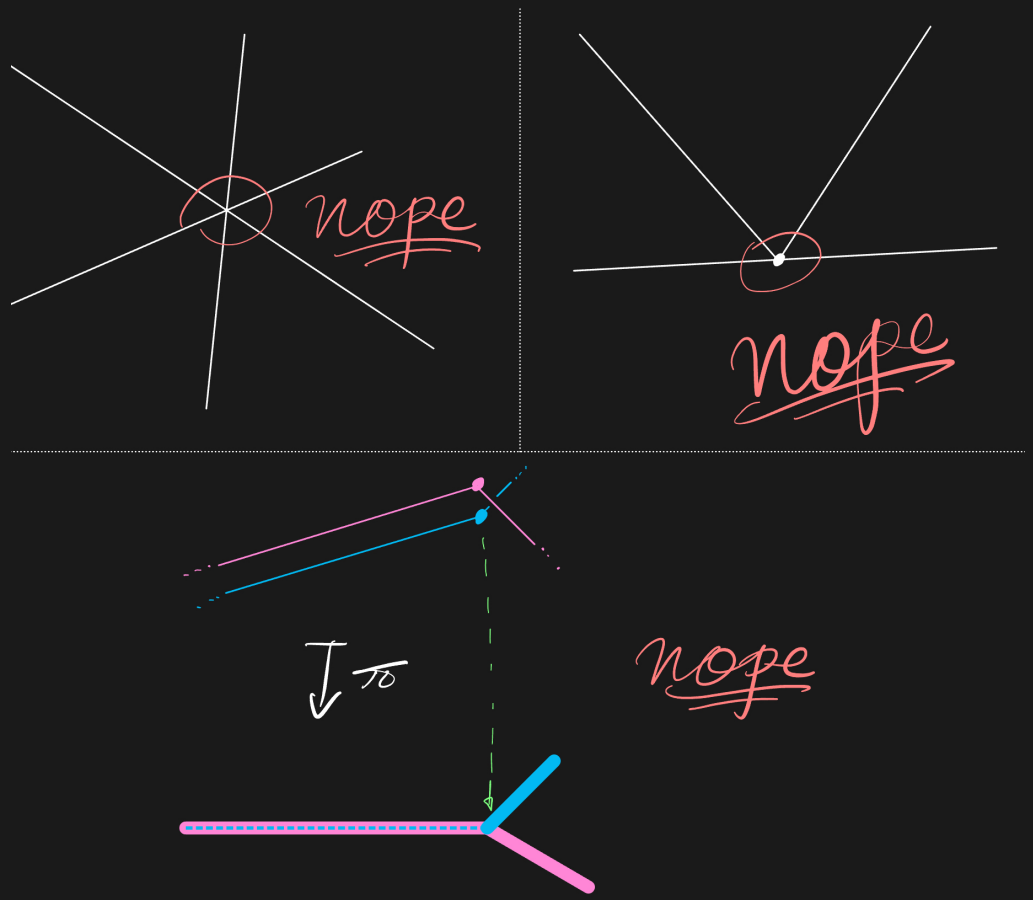
- The *unknotting number* of K is the min. number of "illegal moves" necessary to turn K into the unknot.
- The *knot group* of K is $\pi_1\left(\mathbb{R}^3 \setminus K\right)$.

Question. How can we build nice invariants?

Knot diagrams

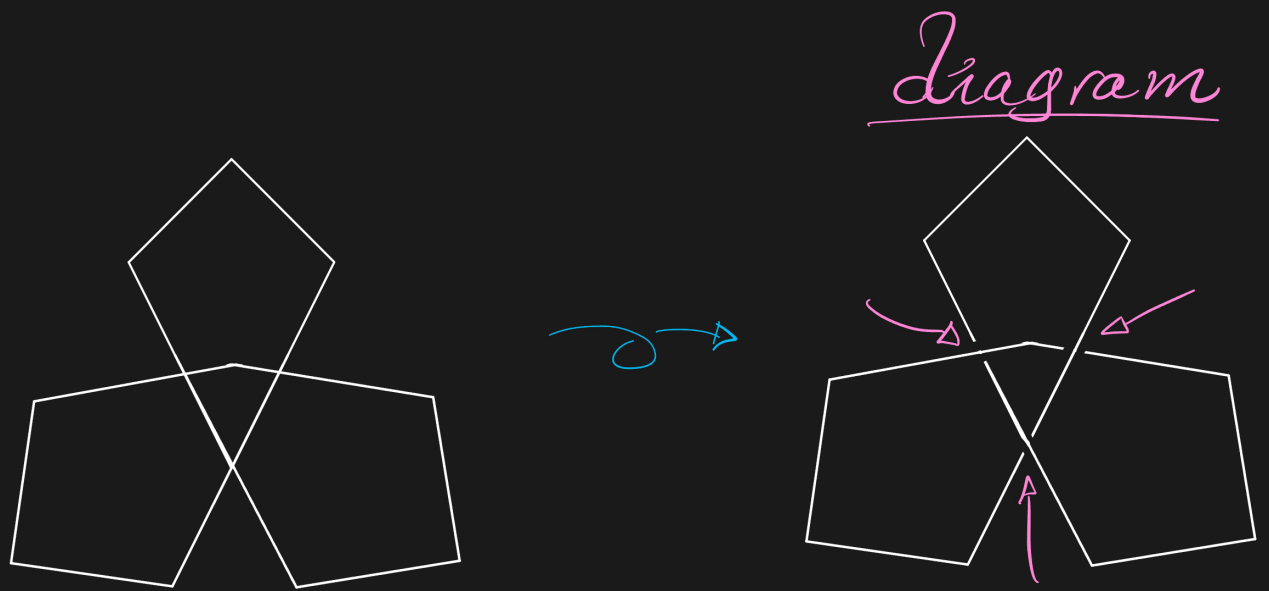
Let $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ project $(x, y, z) \mapsto (x, y)$.

Def. The image of K under π is *regular* just when $\pi^{-1}\{q\}$ meets at most two (closed!) edges of K for all $q \in \mathbb{R}^2$.



Knot diagrams

Def. By attaching "over and under" crossing information to a regular image, we get a *knot diagram*. [4]



Knot diagrams

Promise. Every knot has many diagrams. (We'll return later.)

Knot diagrams

The Δ-moves possible on knots give us the following...

Def. A Δ_1-*move* on a diagram is a Δ-move in $\mathbb{R}^2$.

Knot diagrams

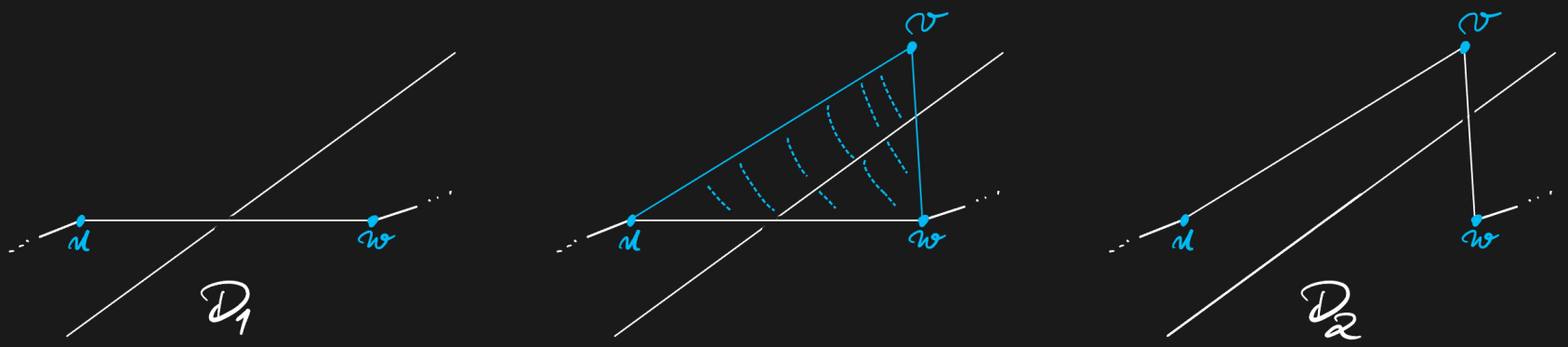
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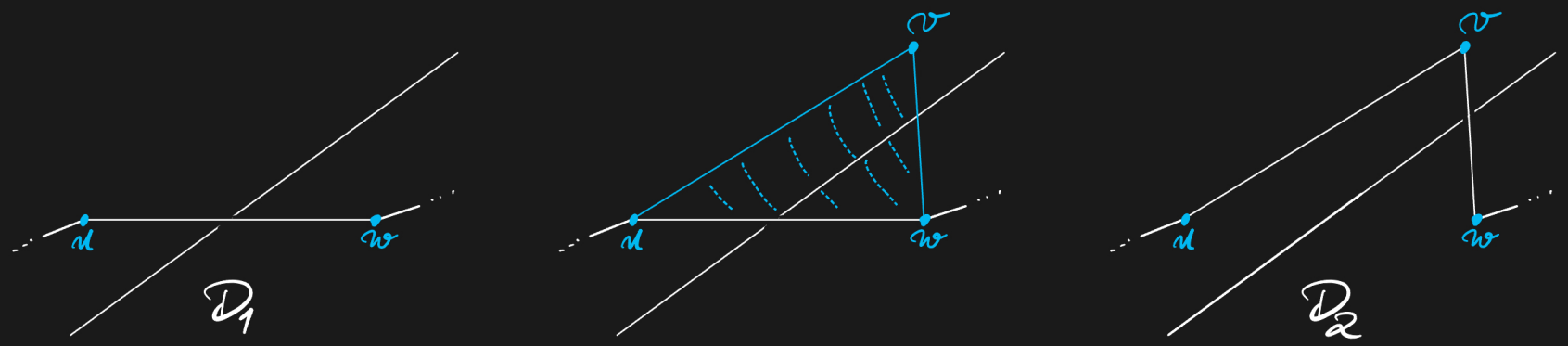
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Def. Δ_i^{-1} are the reverse operations. Two diagrams are *isotopic* if they are related by a sequence of $\Delta_1^{\pm 1}$ and $\Delta_2^{\pm 1}$ moves. [4]

Knot diagrams

Question. Can we define knot invariants using diagrams?

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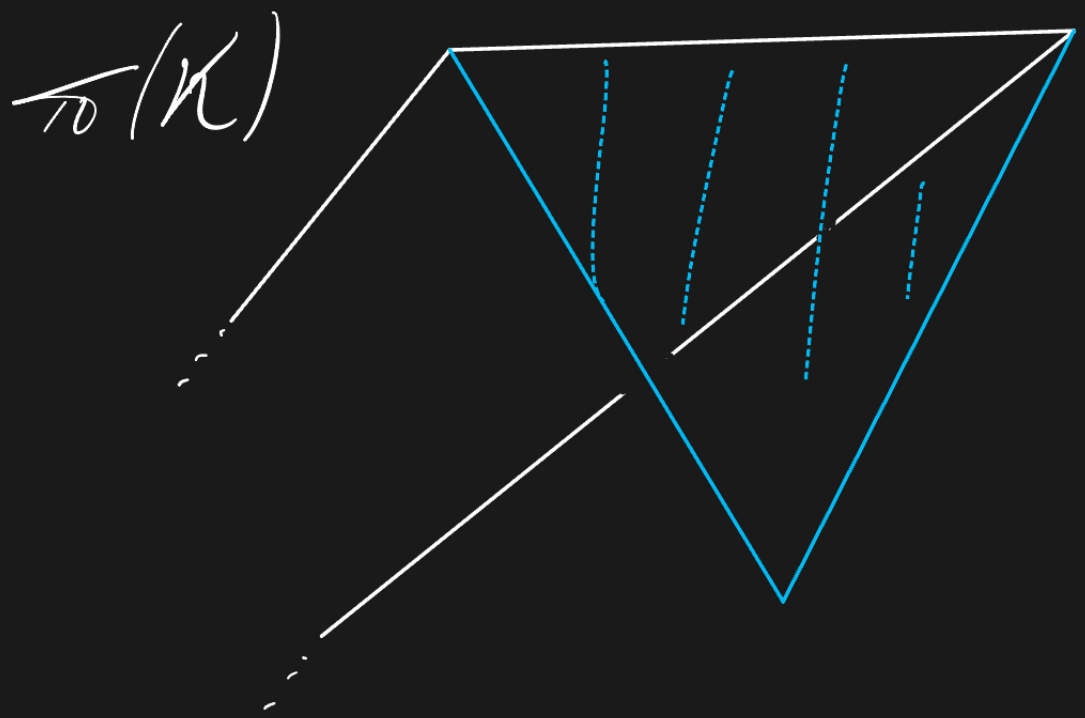
- Sure: the *crossing number* of K is the minimum number of crossings in a diagram of K.

Better question. Can we define invariants computable from *any particular diagram*?

Reidemeister moves

On a diagram $\pi(K)$...

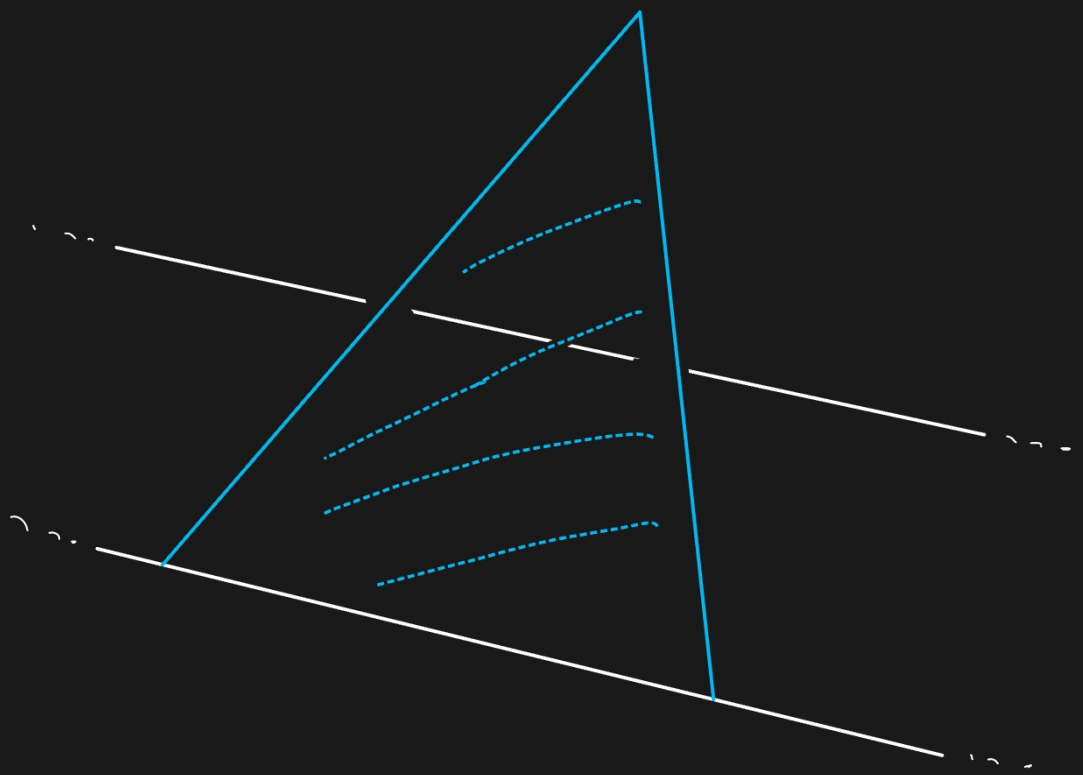
Def. An Ω_1-*move* is the image of a Δ-move of K which turns a projected path with no double points into a single loop.



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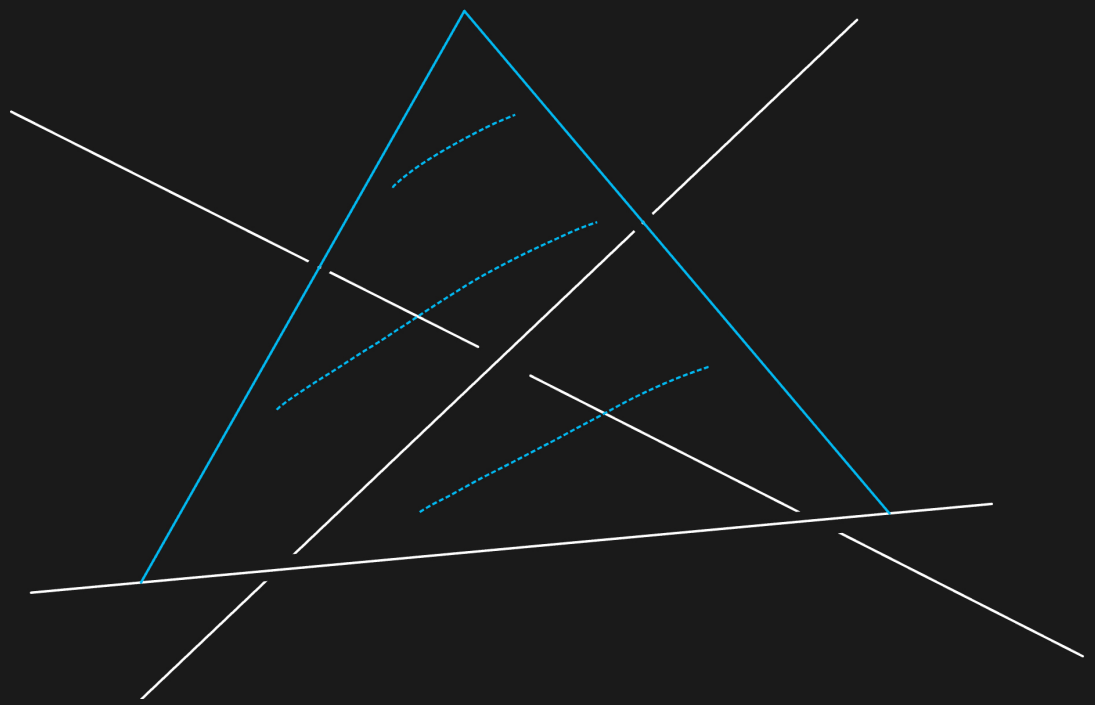
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Theorem (Reidemeister, 1927). Knots modulo isotopy are in bijective correspondence to knot diagrams modulo isotopy and Reidemeister moves.

Original phrasing. *–unintelligible German–* /s
Translated. Any property of knots (mod. isotopy) coincides with properties of diagrams (mod isotopy) preserved under the $\Omega_i^{\pm 1}$ [4].

Consequences of Reidemeister's theorem

- A function defined on diagrams is a knot invariant iff its values are unchanged by diagram isotopy and the Reidemeister moves.

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- The *Jones polynomial* $V : \{\text{oriented links}\} \to \mathbb{Z}[t^{1/2}, t^{-1/2}]$ is an invariant [2].



Proof of Reidemeister's theorem [4]

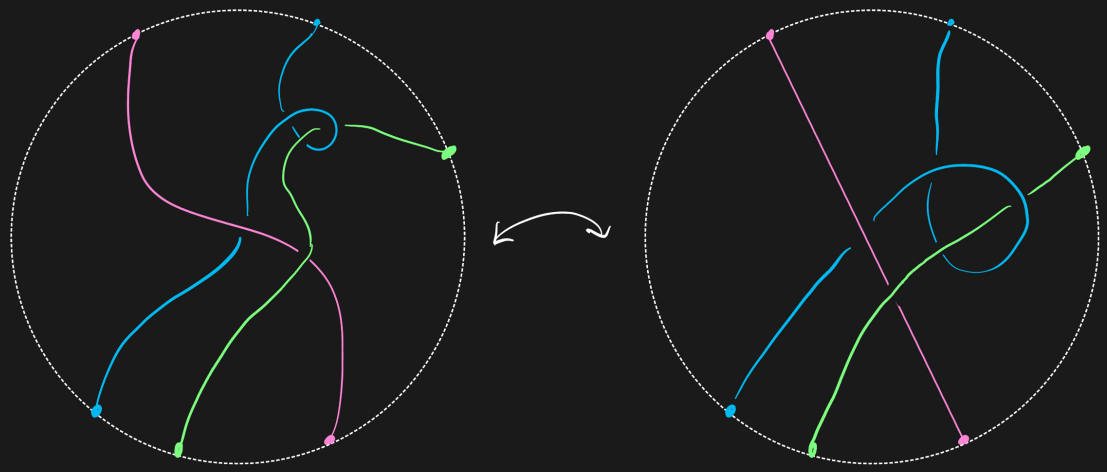
Diagrams $\rightsquigarrow$ knots. Each of $\Delta_1, \Delta_2, \Omega_1, \Omega_2, \Omega_3$ are by definition the image of a Δ move.

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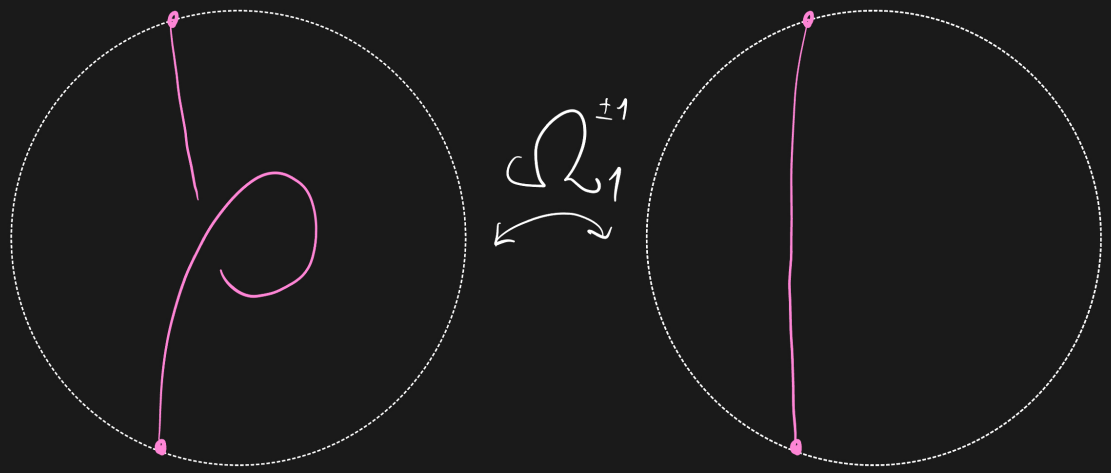


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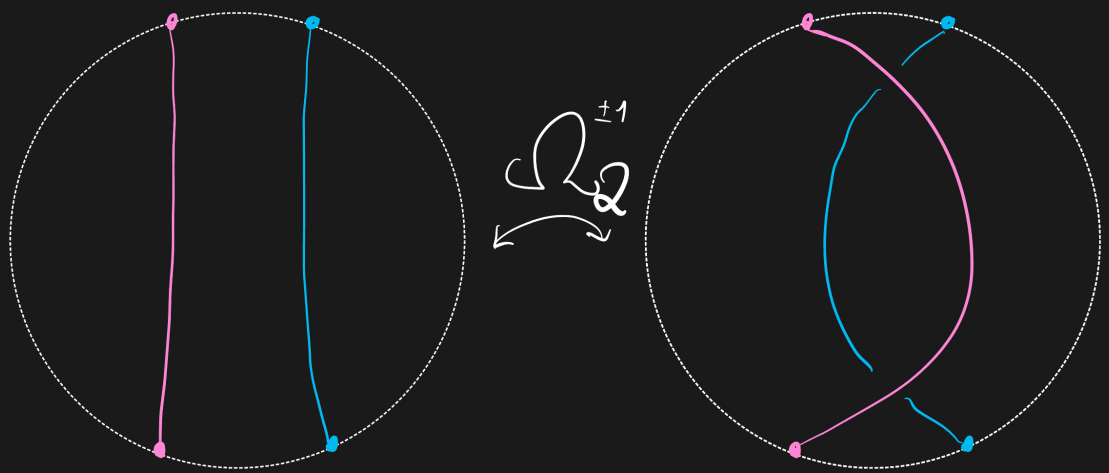


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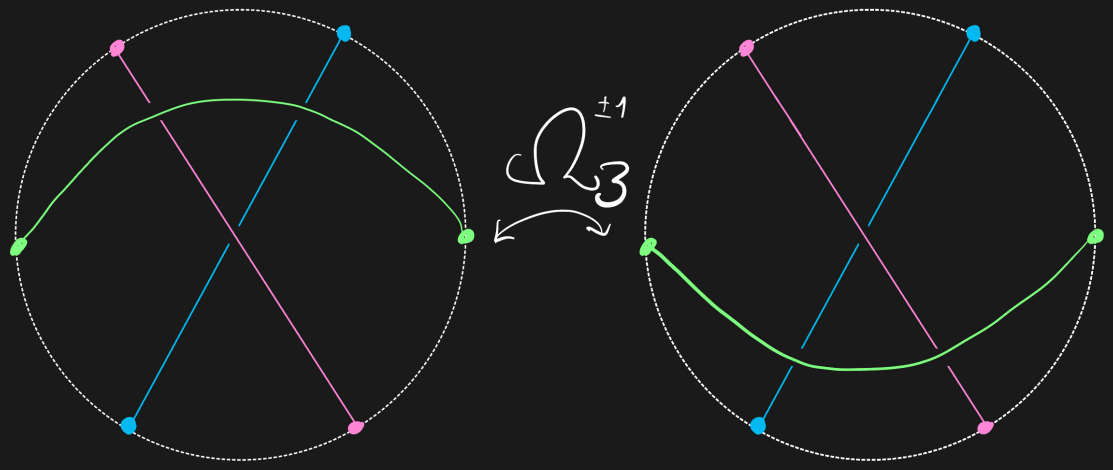


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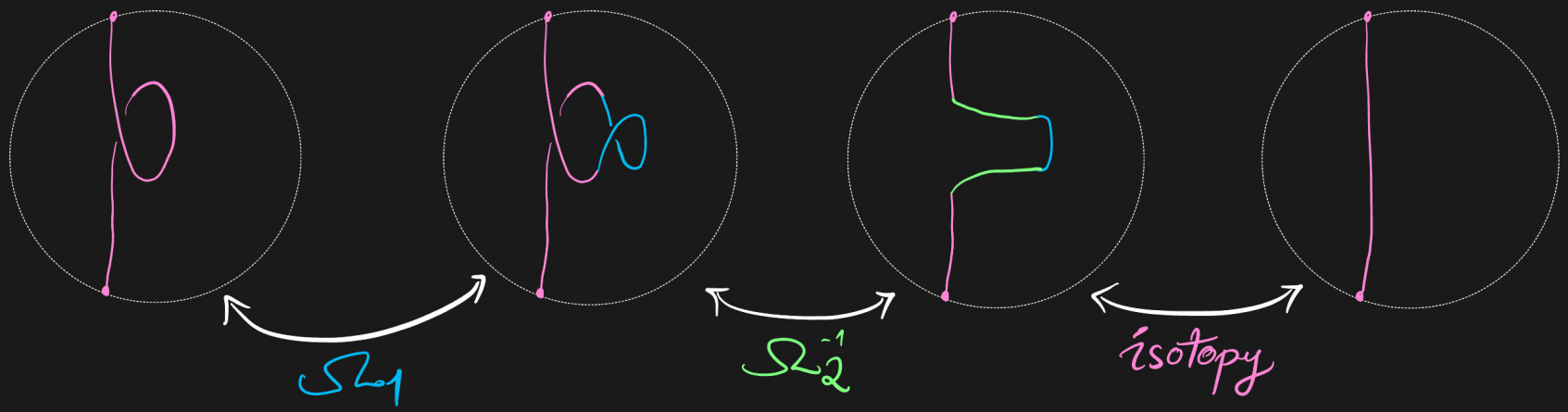


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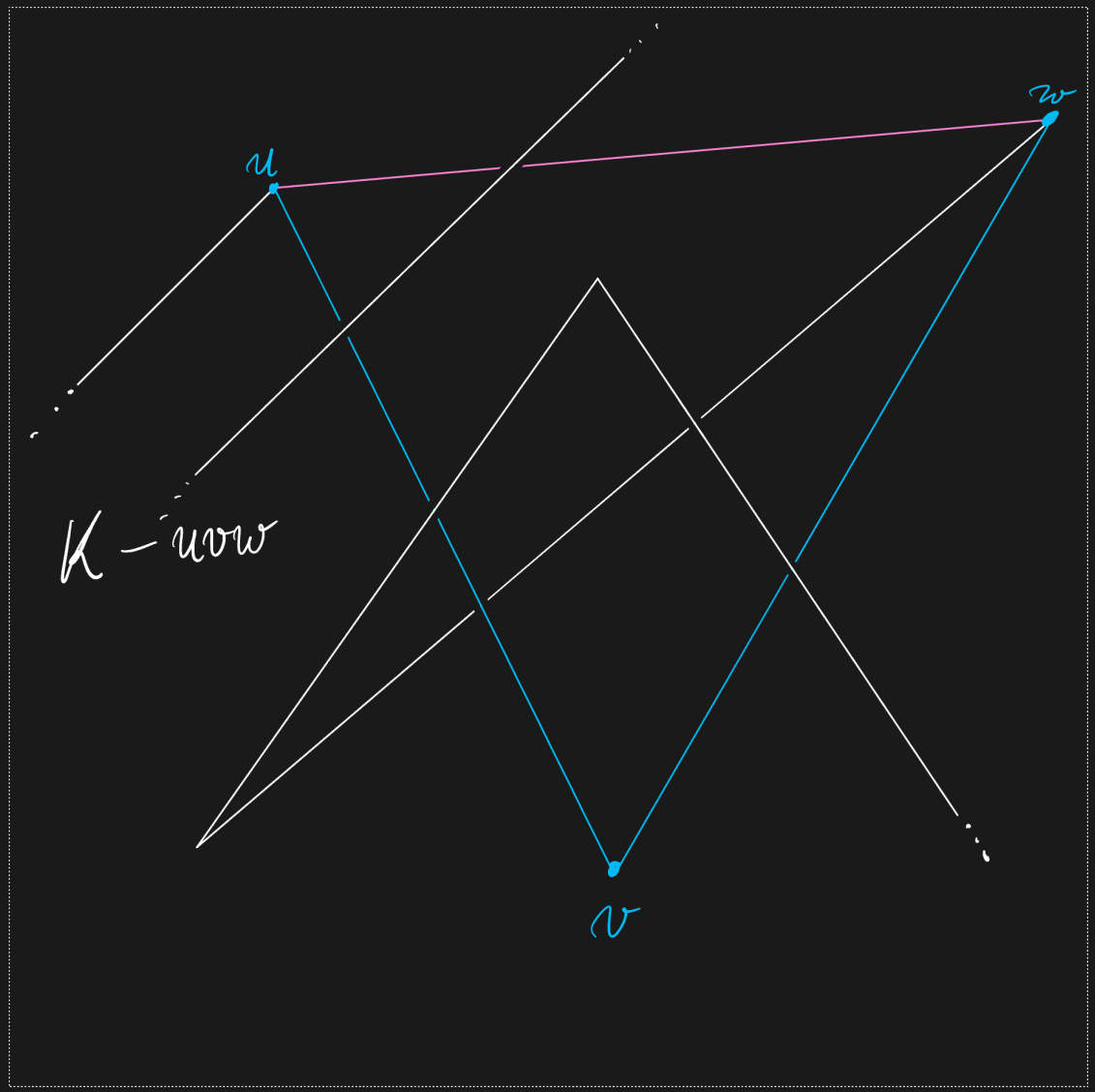
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- Don't mention $\Delta_i^{\pm 1}$; denote all $\Omega_i^{\pm 1}$ "versions" by Ω_i.
- It's enough to recreate the effect of an arbitrary Δ-move on K upstairs by Ω_i-moves downstairs.

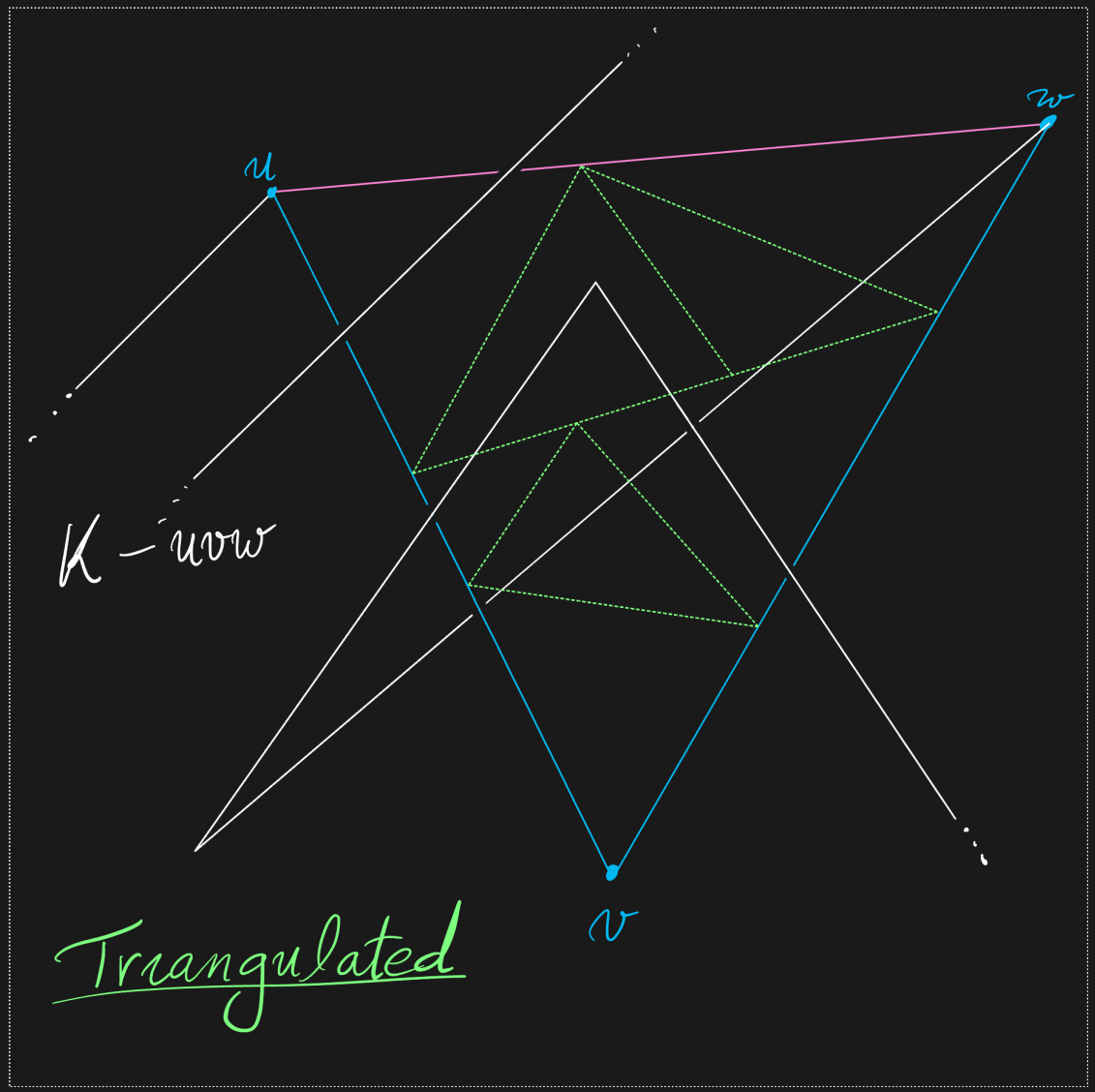
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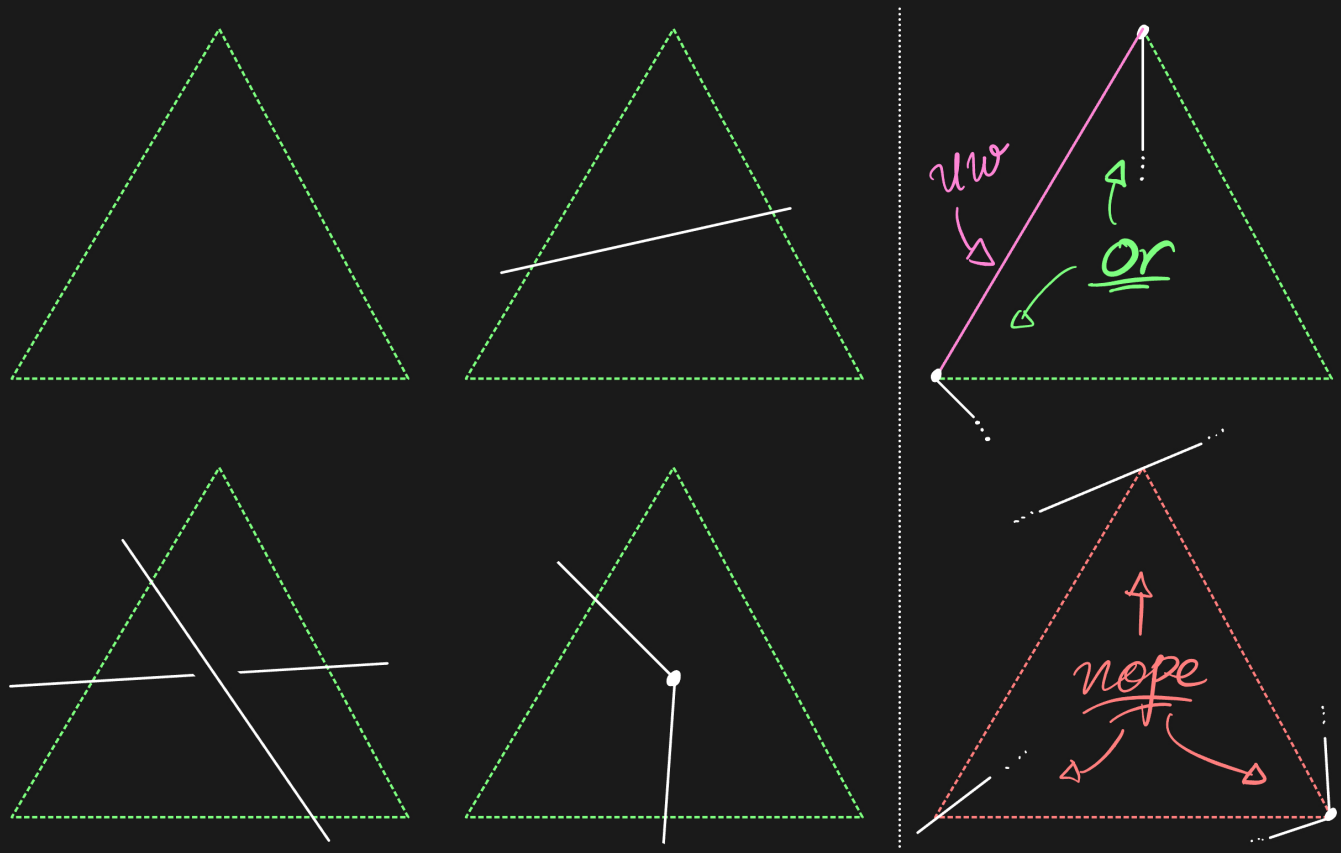
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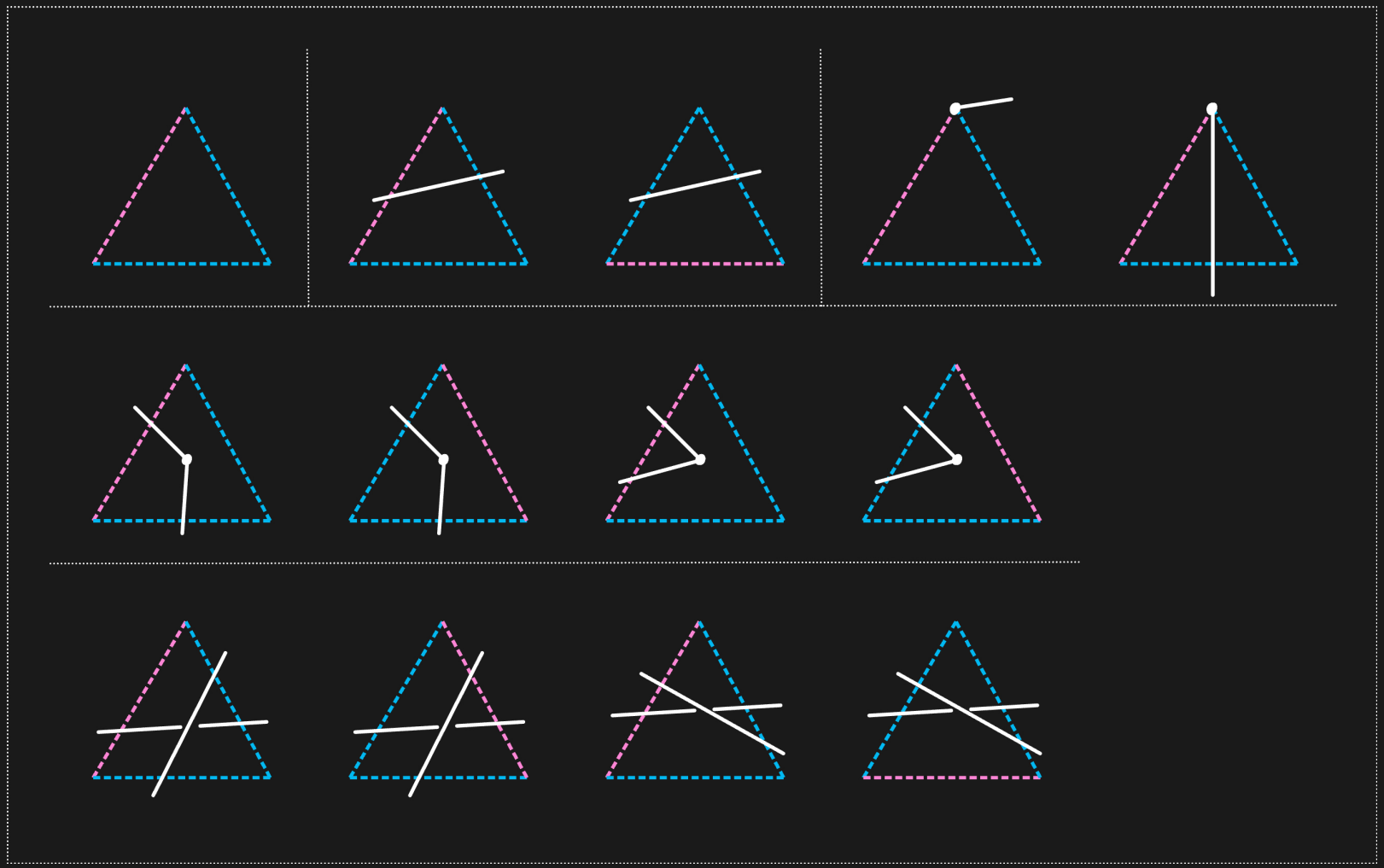
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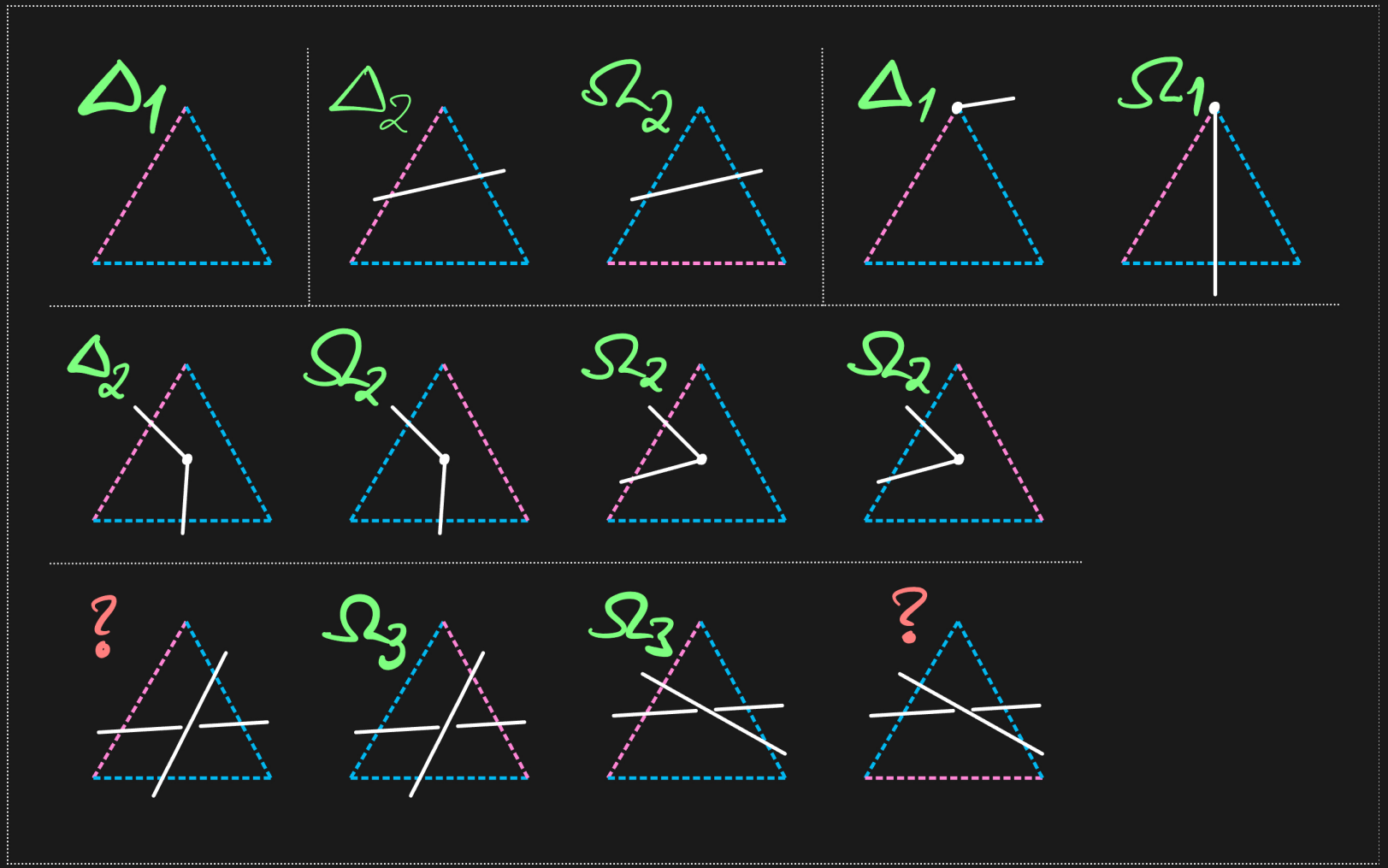
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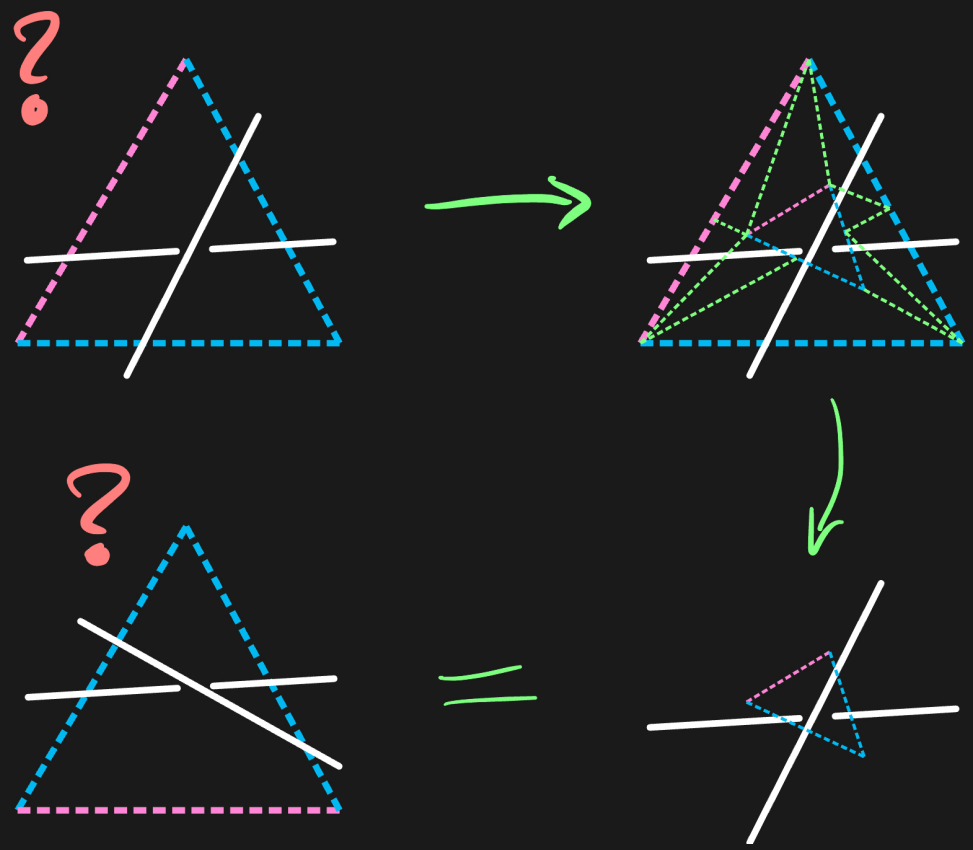
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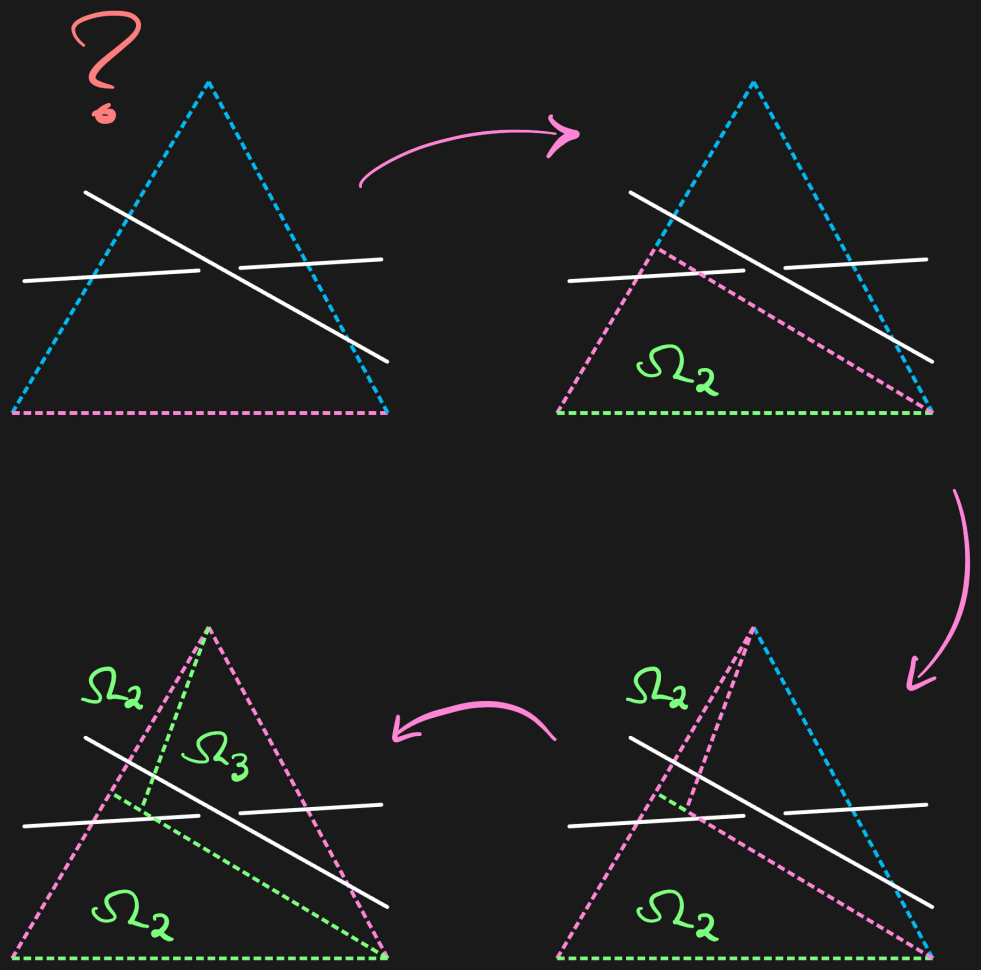
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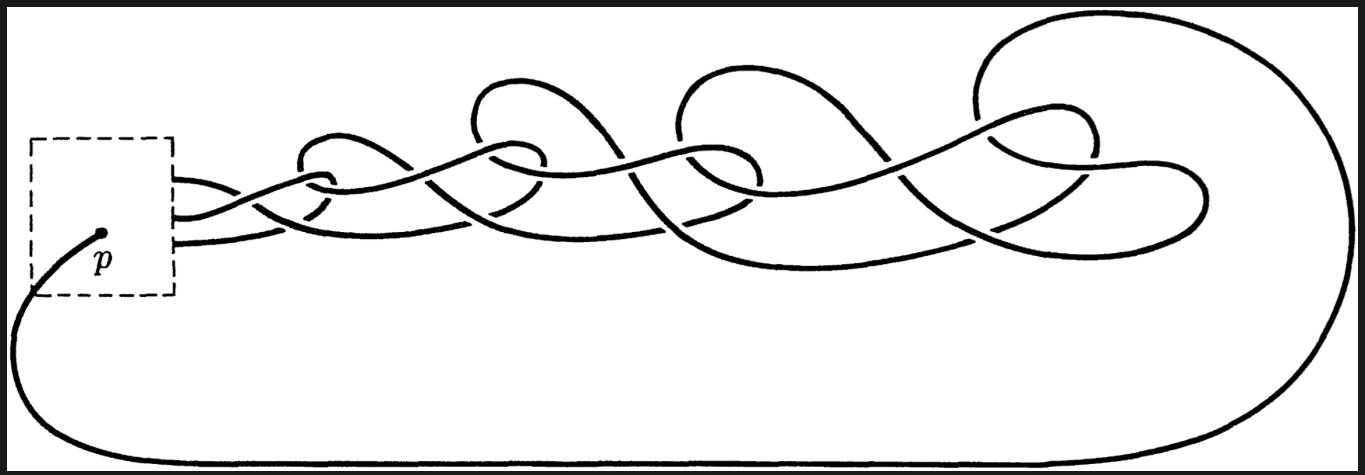


Figure: Fox's *remarkable knot*; figure 5 in [1].

We must focus on *tame knots*.

Smooth knots

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Def. Two knots $f, g : \mathbb{S}^1 \hookrightarrow \mathbb{R}^3$ are *ambient isotopic* just when there is an ambient isotopy h with $g = h_1 \circ f$ and $h_0 = \mathrm{id}$.

Smooth knots

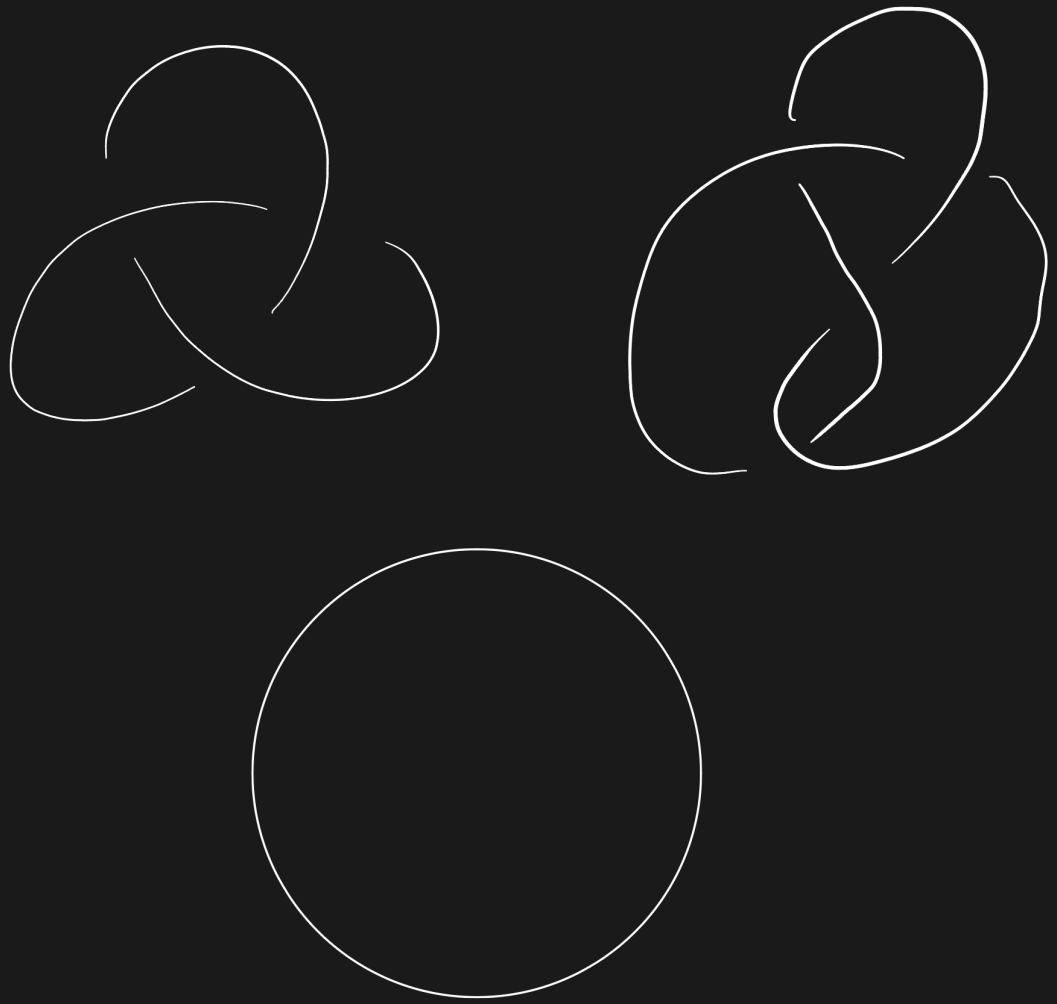
Def. A (tame) *knot* is a smooth injection $\mathbb{S}^1 \hookrightarrow \mathbb{R}^3$.

Def. An *ambient isotopy* is a smooth map $h_\cdot(\cdot) : Y \times \mathbb{I} \to Y$ with $(y, t) \mapsto (h_t(y), t)$ a of diffeomorphism $Y \times \mathbb{I}$.

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Def. A (tame) *knot* is either a curve or a class of them modulo ambient isotopy.

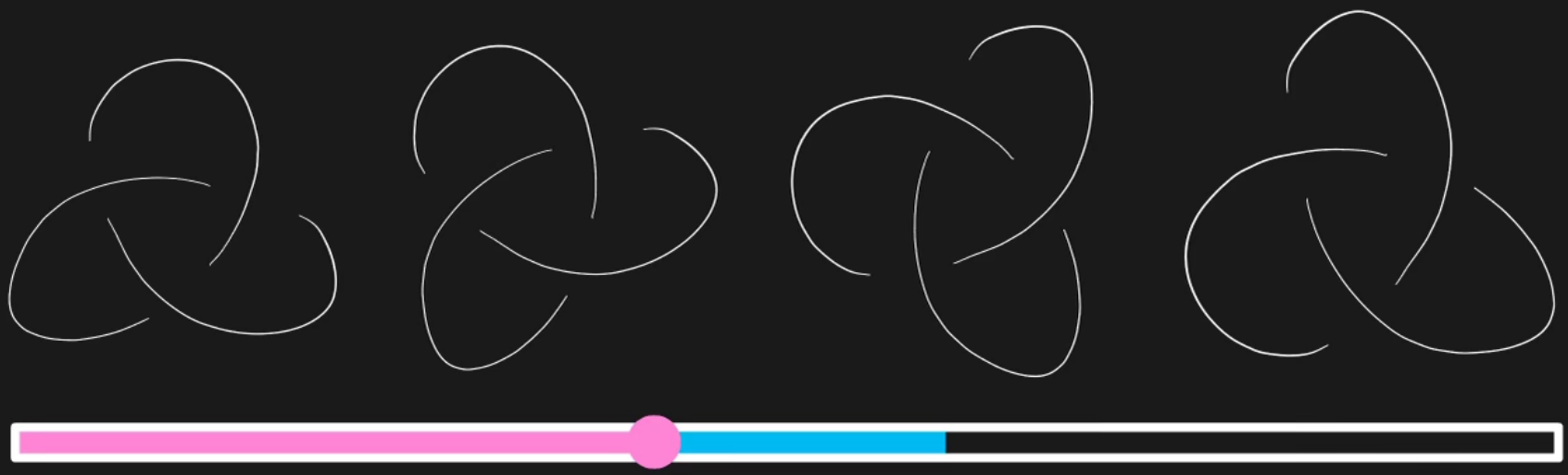
Smooth knots



Examples of ambient isotopies

Isotopies can be:

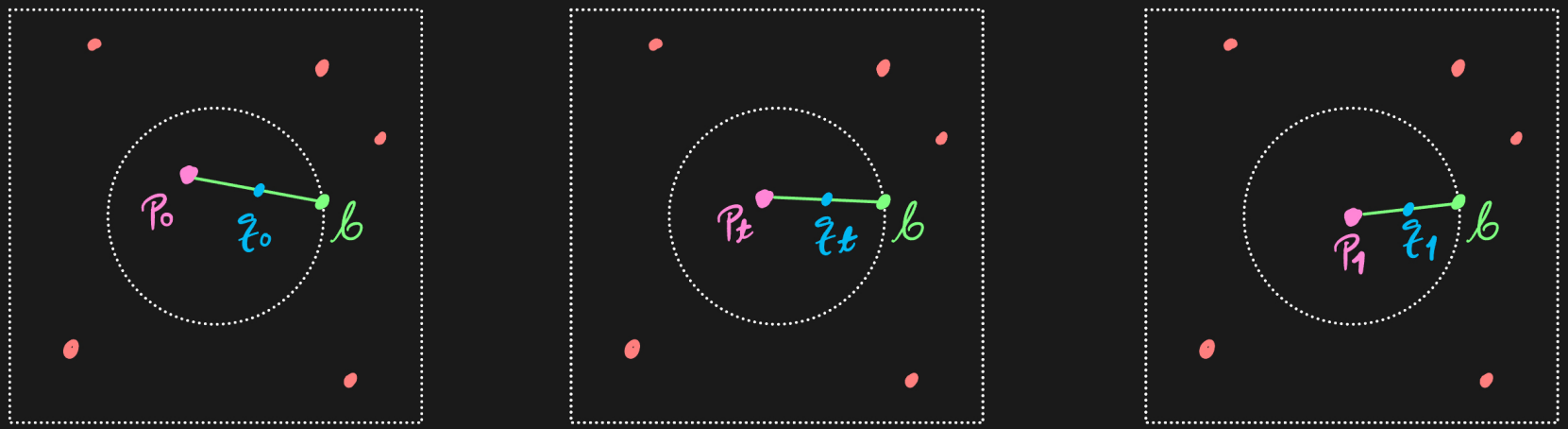
- *global*



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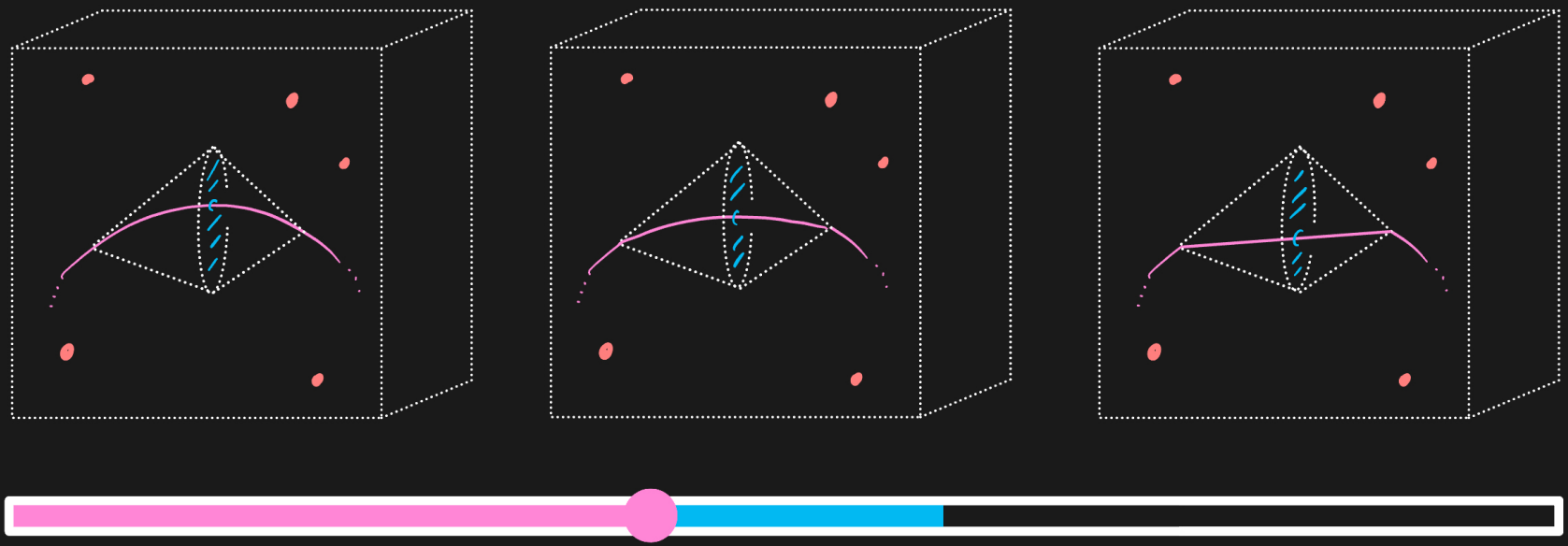
- *global*
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Examples of ambient isotopies

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- *global*
- *local*



See [1] appendix I.

Knot diagrams

Let $\mathbb{S}^1 \overset{f}{\hookrightarrow} \mathbb{R}^3 \overset{\pi}{\twoheadrightarrow} \mathbb{R}^2$ be a knot and the projection.

Def. $\pi \circ f$ is a *knot diagram* just when:

(3) the only multiple points are double points;

(2) double points are transverse;

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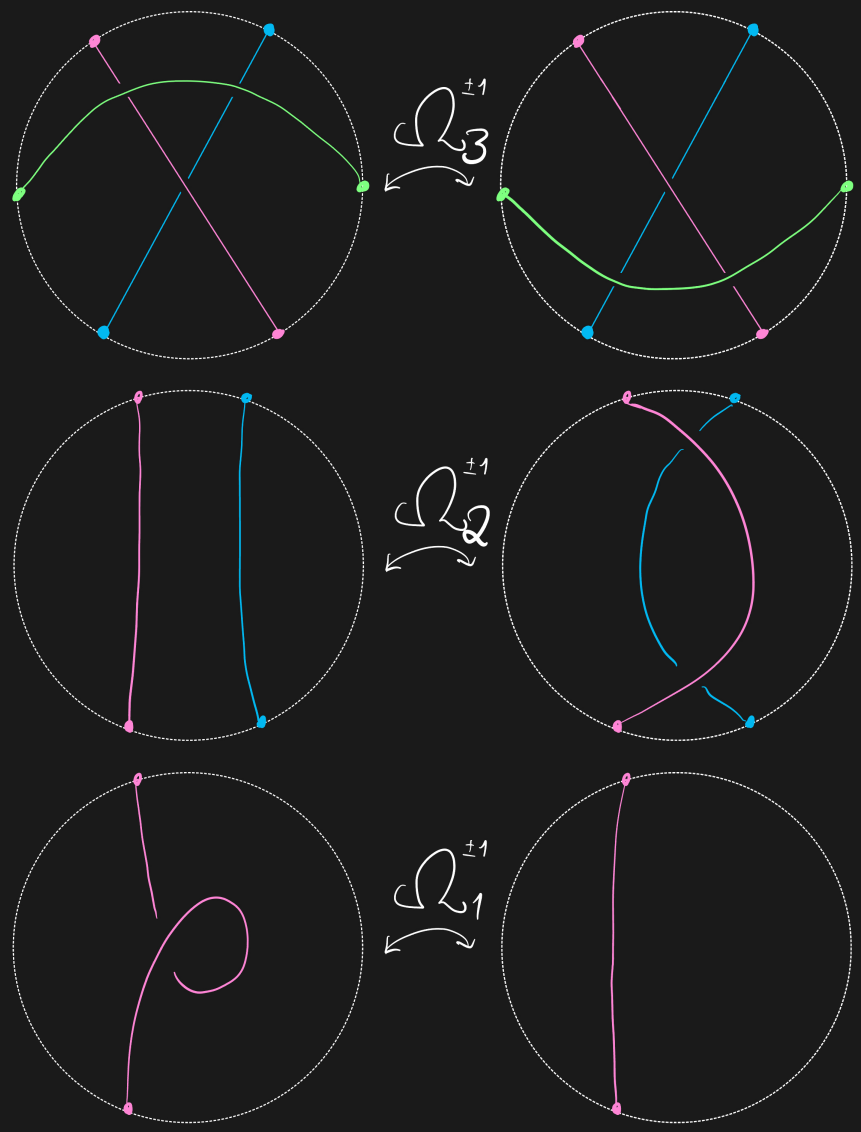
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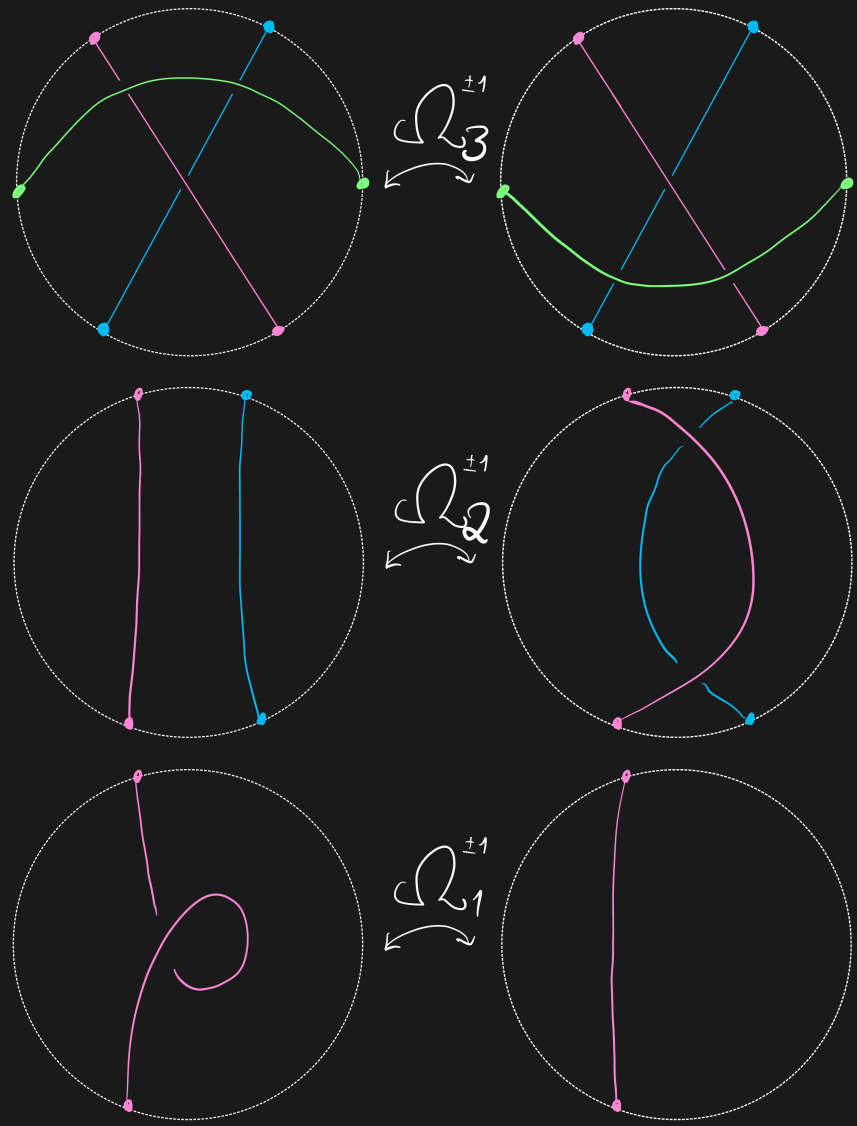
Remark. Two knot diagrams may be ambient isotopic (in $\mathbb{R}^2$).

Reidemeister moves



Reidemeister moves

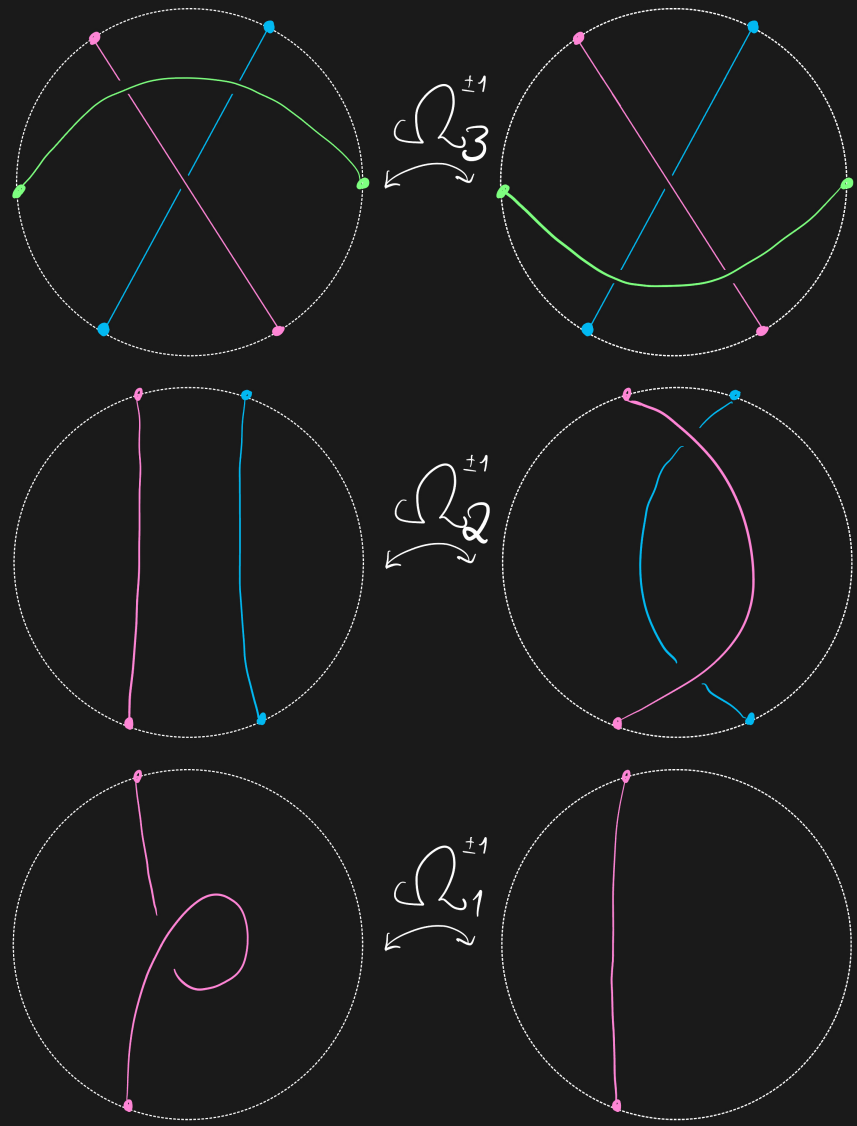
(3) multiple $\Longrightarrow$ double



Reidemeister moves

(3) multiple $\Longrightarrow$ double

(2) double $\Longrightarrow$ transverse

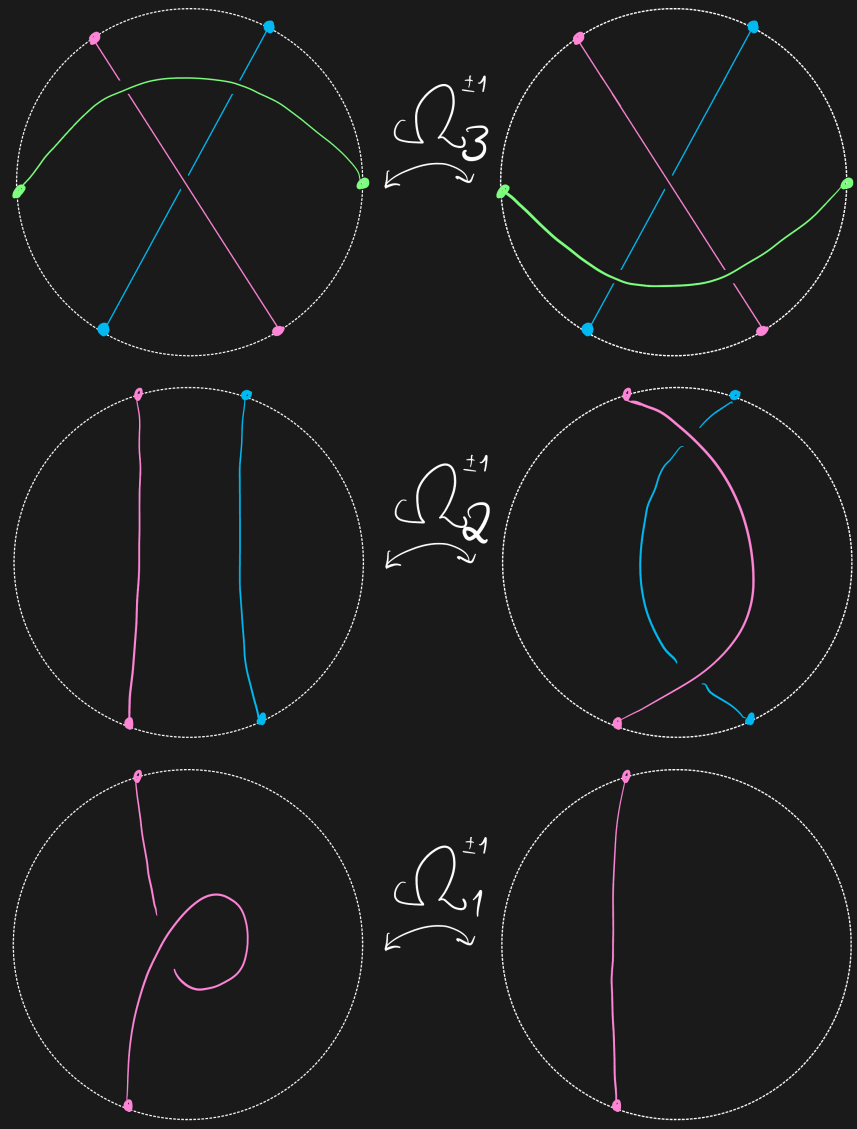


Reidemeister moves

(3) multiple $\Longrightarrow$ double

(2) double $\Longrightarrow$ transverse

(1) $p \circ f' \neq 0$



Reidemeister's theorem

Theorem (Reidemeister, 1927). Knots modulo ambient isotopy are in bijective correspondence to knot diagrams modulo ambient isotopy and Reidemeister moves.

A useful tool in this setting will be *multijet transversality*.

Transversality

Def. Two submanifolds V, W of X are *transverse* just when

$$T_pX = T_pV \oplus T_pW$$

for all $p \in V \cap W$.

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- If $\dim(V) < \operatorname{codim}(W)$ and they are transverse, then $V \cap W = \varnothing$.
- If $\dim(V) = \operatorname{codim}(W)$ and they are transverse, then $V \cap W$ is discrete.

Multijet bundles [3]

Loose def. The *n-fold k-jet bundle* $J^k_n(X, Y)$ describes the possible behaviours up to order k of smooth functions $X \to Y$ at n-tuples of distinct points in X.

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Loose def. Given $f \in C^\infty(X,Y)$, the map $j^k_n f$ sends n-tuples $(x_i)_{i=1}^n$ of distinct points in X to the k-th order behaviour of f at these points. Its image is called the *graph* of f, denoted $\mathrm{Graph}(f)$.

$$f : \mathbb{R}^1 \to \mathbb{R}^3$$

$$\mathrm{Im}(j^0_1 f) = \{(x, f(x)) \mid x \in X\}$$

$$\mathrm{Im}(j^1_3 f) = \left\{(x_i, f(x_i), f'(x_i))_{i=1}^3 \;\middle|\; x_i \in \mathbb{R} \text{ are distinct}\right\}$$

Multijet bundles [3]

Let $W \subseteq J_2^0(\mathbb{S}^1, \mathbb{R}^3)$ consist of points $(\theta_1, p_1, \theta_2, p_2)$ so that $\theta_1 \neq \theta_2$ but $p_1 = p_2$.

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A point of $\mathrm{Graph}(f) \cap W$ consists of

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Points of $\mathrm{Graph}(f) \cap W$ are *failures of injectivity* of $f : \mathbb{S}^1 \to \mathbb{R}^3$.

Multijet transversality [3]

Theorem (Mather, 1970). For W a submanifold of $J^k_n(X, Y)$,

$$\{f \in C^\infty(X, Y) \mid \mathrm{Graph}(f), W \text{ transverse}\}$$

is dense in $C^\infty(X, Y)$.

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Theorem (Mather, 1970). For W a submanifold of $J^k_n(X, Y)$, generic smooth maps have graphs transverse to W. (Lazy interpretation.)

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Result. The projection $\pi \circ f$ of a knot f is generically a knot diagram. (See references.)

Recovering Reidemeister's theorem [3]

Analyse conditions (3), (2), (1) for the composite $f : \mathbb{S}^1 \times \mathbb{I} \to \mathbb{R}^3$ of a knot with an ambient isotopy.

Recovering Reidemeister's theorem [3]

Recall. (3) The only multiple points are double points.

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For generic $\mathrm{Graph}(f) \subseteq J_3^0$,

- $\mathrm{Graph}(f)$ has dimension 6;
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so $\mathrm{Graph}(f) \cap W$ *is finite*.

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Elements of $\mathrm{Graph}(f) \cap W$ describe *triple (or higher) points*.

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For generic $f \in C^\infty(\mathbb{S}^1 \times \mathbb{I}, \mathbb{R}^3)$,

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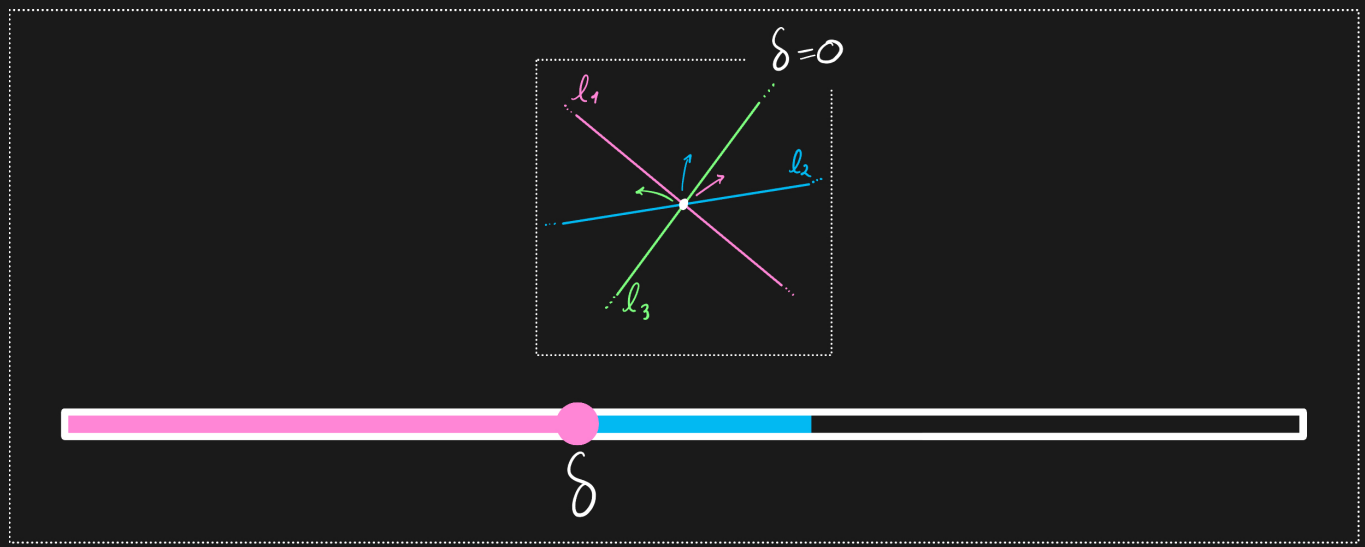
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View $(\pi \circ f)(\theta_i + \varepsilon, t + \delta)$:



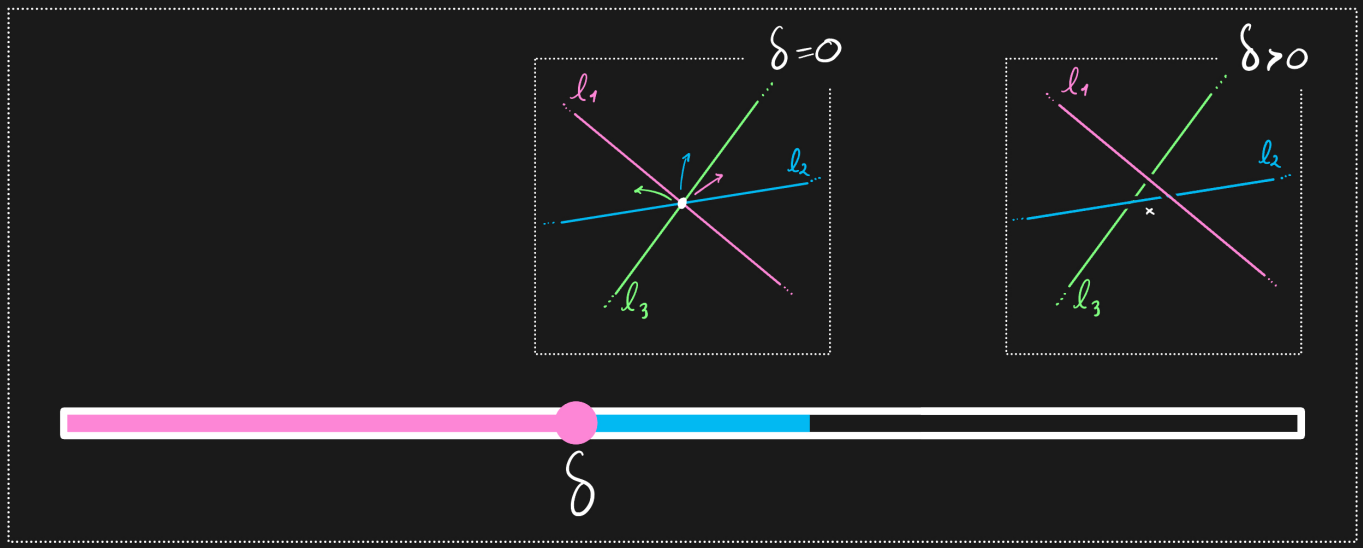
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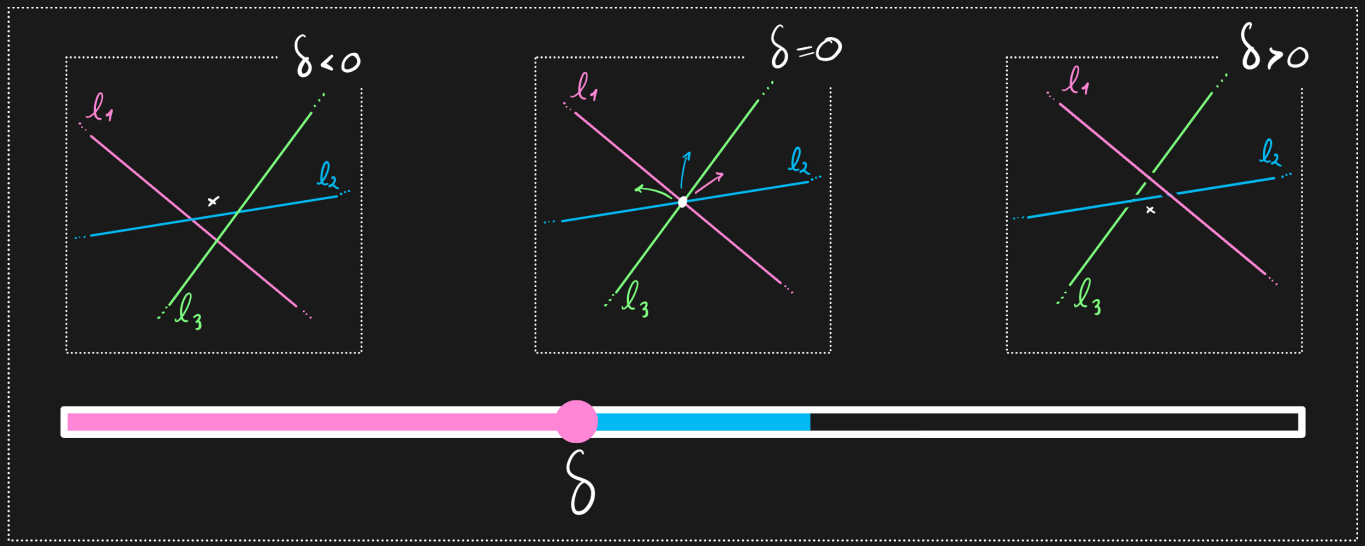
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This is Ω_3.

Recovering Reidemeister's theorem [3]

Recall. (2) Double points are transverse.

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$$\subseteq J_2^1(\mathbb{S}^1 \times \mathbb{I}, \mathbb{R}^3)$$

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- $\mathrm{Graph}(f) \cap W$ is finite, describing non-transverse double points.
- Not secretly triple(-or-more) points.

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- $\mathrm{Graph}(f) \cap W$ is finite, describing non-transverse double points.
- Not secretly triple(-or-more) points.
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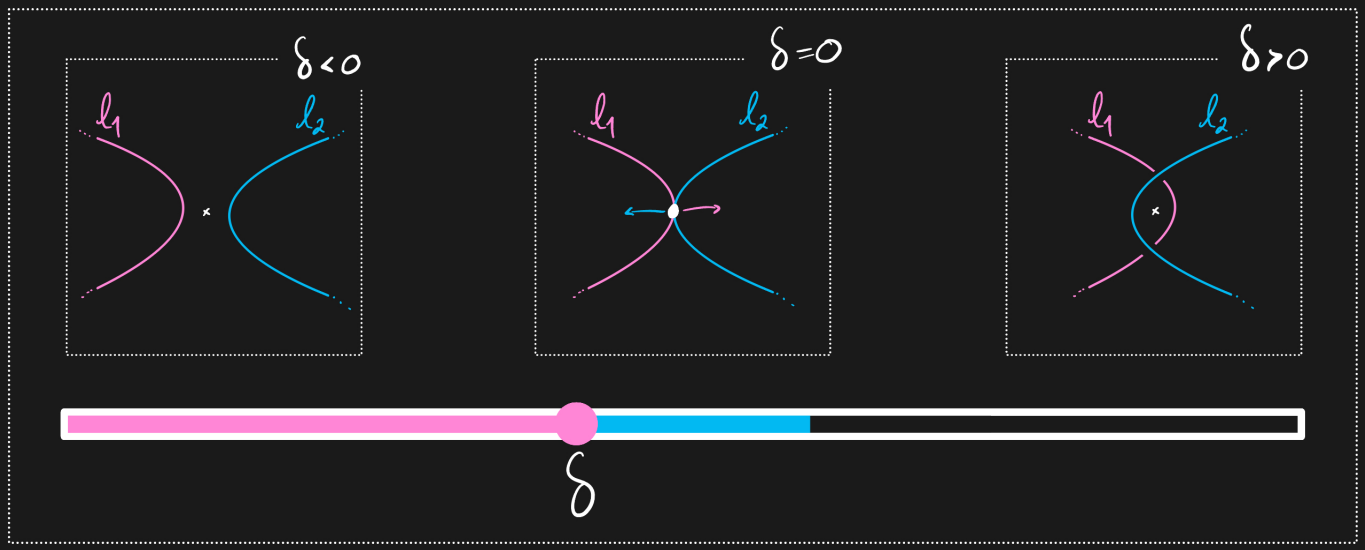
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- $(\pi \circ \partial_t f)(\theta_i, t)$ not collinear.
- $\pi \circ \partial_{\theta\theta} f$ differs.

Recovering Reidemeister's theorem [3]

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This is Ω_2.

Recovering Reidemeister's theorem [3]

Recall. (1) $\pi \circ \partial_\theta f$ never vanishes.

Recovering Reidemeister's theorem [3]

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$$W := \left\{ (\theta, t, p, p^\theta, p^t) \;\middle|\; \pi(p^\theta) = 0 \right\}$$
$$\subseteq J^1_1(\mathbb{S}^1 \times \mathbb{I}, \mathbb{R}^3)$$

Recovering Reidemeister's theorem [3]

Recall. (1) $\pi \circ \partial_\theta f$ never vanishes.

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$$\subseteq J^1_1(\mathbb{S}^1 \times \mathbb{I}, \mathbb{R}^3)$$

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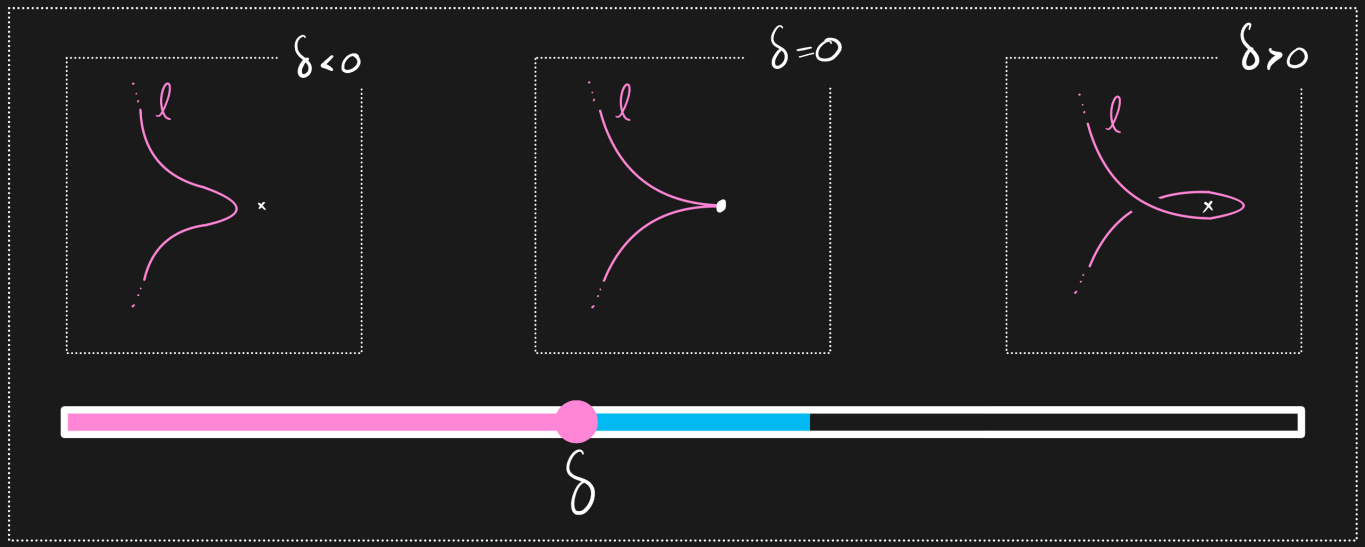
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- $\mathrm{Graph}(f) \cap W$ is finite, describing cusps.
- Not multiple points.
- The correct thing to make the following argument work $(\pi \circ \partial_{\theta\theta\theta} f)$ is nonzero.

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This is Ω_1.

Recovering Reidemeister's theorem [3]

In the projection $\pi \circ f$:

- the moves Ω_i occur finitely many times each,
- in-between, $\pi \circ f$ is deformed by ambient isotopy in $\mathbb{R}^2$.

This is Reidemeister's theorem.



Outline

1 The Polygonal Perspective

2 The Smooth Perspective

3 Summary

Tying loose ends

$$
\begin{aligned}
& \text{piecewise linear curves } \mathbb{S}^1 \hookrightarrow \mathbb{R}^3 \\
\iff & \text{diagrams mod. Reidemeister moves} \\
\iff & \text{smooth curves } \mathbb{S}^1 \hookrightarrow \mathbb{R}^3
\end{aligned}
$$

There is a direct connection; see [1] appendix I.

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- Diagrams modulo isotopy and Reidemeister moves are the same as knots modulo isotopy.
- "Actually computable" knot invariants may be defined on diagrams.

References

[1] R. H. Crowell and C. Fox. *Introduction to Knot Theory*. Springer-Verlag, 1963. ISBN: 978-1-4612-9935-6. DOI: 10.1007/978-1-4612-9935-6.

[2] W.B. Raymond Lickorish. *An Introduction to Knot Theory*. Springer New York, NY, 1997. ISBN: 978-1-4612-0691-0. DOI: 10.1007/978-1-4612-0691-0.

[3] Hoel Queffelec. *Reidemeister's Theorem Using Transversality*. 2014. URL: https://arxiv.org/abs/2406.18203.

[4] Kurt Reidemeister. *Knotentheorie*. Original 1932 work translated by Leo F. Boron, Charles O. Christenson and Bryan A. Smith. BCS Associates, 1983. ISBN: 0-914351-00-1.

Thanks!

